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Display apparatus having display module and manufacturing method thereof

Abstract

A display module includes: a substrate having a mounting surface, four side surfaces, and a rear surface opposite to the mounting surface, the substrate including a thin film transistor layer (TFT) provided on the mounting surface; a plurality of inorganic light-emitting diodes provided on the mounting surface of the substrate; a side wiring electrically connected to the TFT layer and extending along a first pair of side surfaces among the four side surfaces of the substrate; a front cover covering the TFT layer and the plurality of inorganic light emitting devices in a first direction; a metal plate provided on the rear surface of the substrate; a side cover covering the side wiring and the four side surfaces; and a side member provided on a side of the side cover and grounded to the metal plate, wherein the side member is provided on a first side surface of the first pair of side surfaces along which the side wiring extends among the four side surfaces.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation application of International Application No. PCT/KR2022/021543, filed on Dec. 28, 2022, which is based on and claims priority to Korean Patent Applications No. 10-2022-0014450, filed on Feb. 3, 2022, and Korean Patent Application No. 10-2022-0033440, filed on Mar. 17, 2022, in the Korean Intellectual Property Office, the disclosures of which are incorporated by reference herein in their entireties.

BACKGROUND

1. Field

(1) The disclosure relates to a display apparatus for displaying an image by coupling modules in which self-emissive inorganic light-emitting diodes are mounted on a substrate.

2. Description of Related Art

(2) A display apparatus is an output apparatus for visually displaying images and data information, such as characters, figures, etc.

(3) In general, as a display apparatus, a liquid crystal panel including a backlight or an organic light-emitting diode (OLED) panel configured with an organic compound film itself emitting light in response to current has been widely used. However, the liquid crystal panel has a slow response time and high power consumption, and requires a backlight because it does not emit light by itself. Accordingly, it is difficult to reduce the size of the liquid crystal panel. Also, the OLED panel does not require a backlight and may implement a thin thickness because it emits light by itself.

However, the OLED panel is vulnerable to a burn-in phenomenon in which, when the OLED panel displays the same screen for a long time, a certain area of the screen remains the same even after the screen changes to another screen due to short life cycles of sub pixels. For these reasons, as a new panel for replacing the light crystal panel and the OLED panel, a micro light-emitting diode

(micro LED or μ LED) panel in which inorganic light-emitting diodes are mounted on a substrate and the inorganic light-emitting diodes themselves are used as pixels is being studied.

(4) The micro light-emitting diode panel (hereinafter, referred to as a micro LED panel) is a flat display panel is configured with a plurality of inorganic LEDs each having a size of 100 micrometers or less.

(5) The micro LED panel does not cause the burn-in phenomenon of OLEDs as inorganic light-emitting diodes although the micro LED panel is a self-emissive light-emitting device, and the micro LED panel is excellent in view of brightness, resolution, consumption power, and durability.

(6) The micro LED panel provides higher contrast, a faster response time, and higher energy efficiency compared to LCD panels requiring backlights. Micro LEDs have better brightness, light-emitting efficiency, and life cycle than OLEDs, although both the organic LEDs and the micro LEDs as inorganic light-emitting diodes have high energy efficiency.

(7) Also, by arranging LEDs in units of pixels on a circuit board, substrate level display modularization is possible, and various resolutions and screen sizes may be implemented according to consumers' orders.

SUMMARY

(8) Provided are a display apparatus and a manufacturing method thereof, and provide a technical feature related to electrostatic protection for a display module suitable for enlargement and a display apparatus including the display module.

(9) Additional aspects of the disclosure will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments.

(10) According to an aspect of the disclosure, a display module includes: a substrate having a mounting surface on which a plurality of inorganic light-emitting diodes are mounted and a TFT layer is formed, four side surfaces, and a rear surface being opposite to the mounting surface; a side wiring electrically connected to the TFT layer and extending along a first pair of side surfaces among the four side surfaces of the substrate; a front cover covering the TFT layer and the plurality of inorganic light emitting devices in a first direction; a metal plate provided on the rear surface of the substrate; a side cover covering the side wiring and the four side surfaces; and a side member provided on a side of the side cover and grounded to the metal plate, wherein the side member is provided on a first side surface of the first pair of side surfaces along which the side wiring extends among the four side surfaces.

(11) The side member may include a first side member provided on the first side surface of the first pair of side surfaces along which the side wiring extends, and a second side member provided on a second side surface of a second pair of side surface along which the side wiring does not extend among the four side surfaces.

(12) The first side member and the second side member may be integrally formed as one body and extend along the first side surface and the second side surface.

(13) The side wiring may extend along the four side surfaces, and the side member is provided on the first side surface of the first pair of side surfaces and a second side surface of a second pair of side surfaces among the four side surfaces.

(14) The side member positioned on the first side surface and the second side surface may be integrally formed as one body.

(15) A conductivity of the side member may be is greater than a conductivity of the side cover.

(16) A color of the side member may be black.

(17) The side member may include a metal material.

(18) According to an aspect of the disclosure a display apparatus includes: a plurality of display modules are arranged in an $M \times N$ matrix form; and a frame configured to support the plurality of display modules, wherein each display module of the plurality of display modules includes: a substrate having a mounting surface, four side surfaces, and a rear surface opposite to the mounting

surface on which a plurality of inorganic light-emitting diodes are mounted and a TFT layer is formed, four side surfaces, and a rear surface being opposite to the mounting surface; a side wiring electrically connected with the TFT layer and extending along a first pair of side surfaces among the four side surfaces; a front cover covering the plurality of inorganic light-emitting diodes and the TFT layer in a first direction; a metal plate provided on the rear surface; a side cover covering the side wiring and the four side surfaces; and a side member provided on a side of the side cover and grounded to the metal plate, and wherein the side member is positioned on a first side surface of the first pair of side surfaces along which the side wiring extends among the four side surfaces.

(19) The plurality of display modules may include a first display module and a second display module that is adjacent to the first display module in a direction in which the side wiring of the first display module is extends, and the side member of the first display module may contact a second side surface of the second display module among the first pair of side surfaces along which the side wiring of the second display module extends, and the side member of the second display module is not provided on the second side surface of the second display module.

(20) The first display module may be positioned such that a second side surface among the first pair of side surfaces along which the side wiring extends is adjacent to an edge of the frame, and the side member is not positioned on the second side surface, and the frame may include a frame side member that surrounds the second side surface of the first display module among the first pair of side surfaces along which the side wiring extends.

(21) The frame side member may extend along the edge of the frame.

(22) The side member may include a first side member provided on the first side surface of the first pair of side surfaces along which the side wiring extends, and a second side member provided on a second side surface of a second pair of side surface along which the side wiring does not extend among the four side surfaces.

(23) The first side member and the second side member may be integrally formed as one body and extend along the first side surface and the second side.

(24) Each display module of the plurality of display modules may be positioned such that side surfaces on which neither the first side member nor the second side member are positioned, among the four side surfaces, are respectively adjacent to edges of the frame, and the frame may include a frame side member that surrounds the side surfaces of the each display module of the plurality of display modules, on which neither the first side member nor the second side member are provided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other aspects, features, and advantages of certain embodiments of the present disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

(2) FIG. 1 shows a display apparatus according to an embodiment of the disclosure;

(3) FIG. 2 is an exploded view showing main components of the display apparatus of FIG. 1;

(4) FIG. 3 is an enlarged cross-sectional view of some components of a display module shown in FIG. 1;

(5) FIG. 4 is a rear perspective view of a display module of the display apparatus shown in FIG. 1;

(6) FIG. 5 is a perspective view of some components of the display module shown in FIG. 1;

(7) FIG. 6 is a cross-sectional view showing some components of the display apparatus of FIG. 1 in a third direction;

(8) FIG. 7 is an enlarged cross-sectional view of the some components shown in FIG. 6;

(9) FIG. 8 schematically shows a front side of some components of the display apparatus of FIG. 1;

(10) FIG. 9 is a perspective view showing some components of a display module according to

another embodiment of the disclosure;

(11) FIG. **10** is a cross-sectional view showing some components of a display apparatus according to another embodiment of the disclosure in a second direction;

(12) FIG. **11** is an enlarged cross-sectional view of the some components shown in FIG. **10**; and

(13) FIG. **12** schematically shows a front side of some components of a display apparatus according to another embodiment of the disclosure.

DETAILED DESCRIPTION

(14) Embodiments described in the present specification are only examples of the disclosure, and thus it is to be understood that various equivalents or modified examples, are included in the scope of present disclosure.

(15) It is to be understood that the singular forms ‘a,’ ‘an,’ and ‘the’ used in the following description include plural referents unless the context clearly dictates otherwise. In the drawings, for easy understanding, the shapes, sizes, etc. of components are more or less exaggeratedly shown.

(16) It will be understood that when the terms ‘includes,’ ‘comprises,’ ‘including,’ and/or ‘comprising,’ when used in this specification, specify the presence of stated features, figures, steps, operations, components, members, or combinations thereof, but do not preclude the presence or addition of one or more other features, figures, steps, operations, components, members, or combinations thereof.

(17) Also, the meaning of ‘identical’ in the present specification includes being similar in attributes or being similar within a certain range. Also, ‘identical’ means ‘substantially identical’. The meaning of ‘substantially identical’ should be interpreted that a value falling within a manufacturing error range or a value corresponding to a difference falling within a meaningless range with respect to a reference value is included in the range of ‘identical’.

(18) Throughout the disclosure, the expression “at least one of a, b or c” indicates only a, only b, only c, both a and b, both a and c, both b and c, all of a, b, and c, or variations thereof.

(19) Hereinafter, example embodiments of the disclosure will be described with reference to the accompanying drawings.

(20) FIG. **1** shows a display apparatus according to an embodiment of the disclosure, FIG. **2** is an exploded view showing main components of the display apparatus of FIG. **1**, FIG. **3** is an enlarged cross-sectional view of some components of a display module shown in FIG. **1**, FIG. **4** is a rear perspective view of a display module of the display apparatus shown in FIG. **1**, and FIG. **5** is a perspective view of some components of the display module shown in FIG. **1**.

(21) Components of a display apparatus **1** including a plurality of inorganic light-emitting diodes **50** shown in the drawings are components in micro-units having a size of several μm to several hundreds of micrometers (μm), and the sizes of some components (the plurality of inorganic light-emitting diodes **50**, a black matrix **48**, etc.) shown in the drawings are exaggerated for convenience of description.

(22) The display apparatus **1** is a device that displays information, material, data, etc. as characters, figures, graphs, images, etc., and may be implemented as a television (TV), a personal computer (PC), a mobile, a digital signage, etc.

(23) According to an embodiment of the disclosure, as shown in FIGS. **1** and **2**, the display apparatus **1** may include a display panel **20** for displaying an image, a power supply for supplying power to the display panel **20**, a main board **25** for controlling overall operations of the display panel **20**, a frame **15** supporting the display panel **20**, and a rear cover **10** covering a rear surface of the frame **15**.

(24) The display panel **20** may include a plurality of display modules **30A** to **30P**, a driving board for driving the individual display modules **30A** to **30P**, and a Timing Controller (TCON) board for generating timing signals required for controlling the individual display modules **30A** to **30P**.

(25) The rear cover **10** may support the display panel **20**. The rear cover **10** may be mounted on the floor through a stand, or may be mounted on a wall through a hanger.

(26) The plurality of display modules **30A** to **30P** may be arranged in upper to lower directions and left to right directions to be adjacent to each other. The plurality of display modules **30A** to **30P** may be arranged in an $M \times N$ matrix. In the current embodiment, sixteen display modules **30A** to **30P** are arranged in a 4×4 matrix form, but there is no limitation on the number and arrangement of the plurality of display modules **30A** to **30P**.

(27) For example, the plurality of display modules **30A** to **30P** according to an embodiment of the disclosure may include a first display module **30A** positioned at an uppermost and leftmost location. The plurality of display modules **30B** to **30P** may be arrayed in a right direction or a down direction with respect to the first display module **30A**.

(28) The plurality of display modules **30A** to **30P** may be mounted on the frame **15** through various known methods, such as magnetism using a magnet, a mechanical insertion structure, etc. The rear cover **10** may be coupled with a rear side of the frame **15**, and the rear cover **10** may form a rear outer appearance of the display apparatus **1**.

(29) The rear cover **10** may include a metal material. Accordingly, heat generated from the plurality of display modules **30A** to **30P** and the frame **15** may be easily transferred to the rear cover **10** to raise heat-radiating efficiency of the display apparatus **1**.

(30) As such, the display apparatus **1** according to an embodiment of the disclosure may implement a large screen by tiling the plurality of display modules **30A** to **30P**.

(31) Alternatively, each of the plurality of display modules **30A** to **30P** may be applied to a display apparatus. That is, the display modules **30A** to **30P** may be, in unit of a single piece, installed in and applied to a wearable device, a portable device, or various electronic products or electric parts requiring displays. Also, like the embodiment of the disclosure, the display modules **30A** to **30P** may be applied to a display device, such as a PC monitor, a high-resolution TV, a signage, an electronic display, etc., by being assembled and arranged in a matrix form.

(32) The plurality of display modules **30A** to **30P** may have the same configuration. Accordingly, the following description about any one display module will be applied in the same way to all the other display modules.

(33) Hereinafter, because the plurality of display modules **30A** to **30P** have the same configuration, the plurality of display modules **30A** to **30P** will be described based on the first display module **30A**.

(34) That is, to avoid redundant descriptions, the configuration of the plurality of display modules **30A** to **30P** will be described by representatively using a display module **30**, a substrate **40**, and a front cover **70**.

(35) Also, the first display module **30A**, a third display module **30E** neighboring the first display module **30A** in a second direction Y, or a second display module **30B** neighboring the first display module **30A** in a third direction Z among the plurality of display modules **30A** to **30P** will be described as necessary.

(36) Also, a fifth display module **30I** neighboring the third display module **30E** in the second direction Y and being opposite to the first display module **30A** with respect to the third display module **30E**, a seventh display module **30M** neighboring the fifth display module **30I** and being opposite to the third display module **30E** with respect to the fifth display module **30I**, a fourth display module **30C** neighboring the second display module **30B** in the third direction Z and being opposite to the first display module **30A** with respect to the second display module **30B**, and a sixth display module **30D** neighboring the fourth display module **30C** in the third direction Z and being opposite to the second display module **30B** with respect to the fourth display module **30C** will be described as necessary.

(37) The first display module **30A** among the plurality of display modules **30A** to **30P** may be formed, for example, in a quadrangle type. The first display module **30A** may be provided in a rectangle type or a square type.

(38) Accordingly, the first display module **30A** may include edges **31**, **32**, **33**, and **34** positioned in

upper, lower, left, and right directions with respect to a first direction X which is a front direction. (39) As shown in FIG. 3, each of the plurality of display modules 30A to 30P may include a substrate 40 and a plurality of inorganic light-emitting diodes 50 mounted on the substrate 40. The plurality of inorganic light-emitting diodes 50 may be mounted on a mounting surface 41 of the substrate 40 toward the first direction X. In FIG. 3, for convenience of description, a thickness of the substrate 40 in the first direction X is exaggerated.

(40) The substrate 40 may be formed in a quadrangle type. As described above, each of the plurality of display modules 30A to 30P may be provided in a quadrangle type, and the substrate 40 may also be formed in a quadrangle type correspondingly.

(41) The substrate 40 may be provided in a rectangle type or a square type.

(42) Accordingly, the substrate 40 may include four edges E corresponding to the edges 31, 32, 33, and 34 of the first display module 30A, positioned in the upper, lower, left, and right directions with respect to the first direction X which is the front direction (see FIG. 5).

(43) The first display module 30A may include a right edge 31, an upper edge 32, a left edge 33, and a lower edge 34 with respect to the first direction X as the front direction in which a screen of the first display module 30A is displayed. The right edge 31 may be opposite to the left edge 33 in the second direction Y which is a left-right direction, and the upper edge 32 may be opposite to the lower edge 34 in the third direction Z which is an up-down direction.

(44) The substrate 40 may include a substrate body 42, a mounting surface 41 forming one surface of the substrate body 42, a rear surface 43 forming the other surface of the substrate body 42 and being opposite to the mounting surface 41, and a side surface 45 positioned between the mounting surface 41 and the rear surface 43.

(45) The side surface 45 may form a side of the substrate 40 in the second direction Y and the third direction Z that are orthogonal to the first direction X.

(46) The substrate 40 may include a chamfer portion 49 formed between the mounting surface 41 and the side surface 45 and between the rear surface 43 and the side surface 45.

(47) The chamfer portion 49 may prevent the plurality of display modules 30A to 30P from colliding with each other to be damaged upon an arrangement.

(48) Each edge E of the substrate 40 may include the side surface 45 and the chamfer portion 49.

(49) The substrate 40 may include a Thin Film Transistor (TFT) layer 44 formed on the substrate body 42 to drive the inorganic light-emitting diodes 50. The substrate body 42 may include a glass substrate. That is, the substrate 40 may include a Chip On Glass (COG) type substrate. On the substrate 40, a first pad electrode 44a and a second pad electrode 44b may be formed to electrically connect the inorganic light-emitting diodes 50 with the TFT layer 44.

(50) TFTs constituting the TFT layer 44 are not limited to a certain structure or type, and may be configured as various embodiments. That is, the TFTs of the TFT layer 44 according to an embodiment of the disclosure may be implemented as Low Temperature Poly Silicon (LTPS) TFTs, oxide TFTs, Si (poly silicon or a-silicon) TFTs, organic TFTs, or graphene TFTs.

(51) Also, the TFT layer 44 may be replaced with Complementary Metal-Oxide Semiconductor (CMOS) type, n-type MOSFET, or p-type MOSFET transistors in a case in which the substrate body 42 of the substrate 40 is provided as a silicon wafer.

(52) The plurality of inorganic light-emitting diodes 50 may be formed of an inorganic material, and may include inorganic light-emitting diodes each having a size of several μm to tens of μm in width, length, and height. A length of a shorter side of a micro inorganic light-emitting device among the width, length, and height may be 100 μm or less. That is, the inorganic light-emitting diodes 50 may be picked up from a sapphire or silicon wafer and directly transferred onto the substrate 40. The plurality of inorganic light-emitting diodes 50 may be picked up and conveyed through an electrostatic method using an electrostatic head or a stamp method using an elastic polymer material, such as PDMS or silicon, as a head.

(53) Each of the plurality of inorganic light-emitting diodes 50 may be a light-emitting structure

including a n-type semiconductor **58a**, an active layer **58c**, a p-type semiconductor **58b**, a first contact electrode **57a**, and a second contact electrode **57b**.

(54) For example, Any one of the first contact electrode **57a** may be electrically connected with the second contact electrode **57b** and the n-type semiconductors **58a**, and the other one of the first contact electrode **57a** may be electrically connected with the p-type semiconductor **58b**.

(55) The first contact electrode **57a** and the second contact electrode **57b** may be positioned horizontally, and may be a flip chip type arranged toward the same direction (an opposite direction of a light-emitting direction).

(56) Each inorganic light-emitting device **50** may have a light-emitting surface **54** positioned toward the first direction X upon being mounted on the mounting surface **41**, a side surface **55**, and a bottom surface **56** being opposite to the light-emitting surface **54**, wherein the first contact electrode **57a** and the second contact electrode **57b** may be formed on the bottom surface **56**.

(57) That is, the first and second contact electrodes **57a** and **57b** of the inorganic light-emitting device **50** may be positioned on the opposite side of the light-emitting surface **54**, that is, in the opposite direction of the light-emitting direction.

(58) The first and second contact electrodes **57a** and **57b** may face the mounting surface **41** and be electrically connected with the TFT layer **44**, wherein the light-emitting surface **54** from which light is emitted may be positioned in the opposite direction of the direction in which the contact electrodes **57a** and **57b** are positioned.

(59) Accordingly, upon irradiating of light generated from the active layer **58c** in the first direction X through the light-emitting surface **54**, the light may be radiated toward the first direction X without any interference with the first contact electrode **57a** or the second contact electrode **57b**.

(60) That is, the first direction X may be defined as a direction in which the light-emitting surface **54** is positioned to irradiate light.

(61) The first contact electrode **57a** and the second contact electrode **57b** may be electrically connected respectively with the first pad electrode **44a** and the second pad electrode **44b** formed on the mounting surface **41** of the substrate **40**.

(62) The inorganic light-emitting device **50** may be connected directly with the first and second pad electrodes **44a** and **44b** through a bonding component, such as an anisotropic conductive layer **47** or a solder.

(63) The anisotropic conductive layer **47** may be formed on the substrate **40** to mediate electrical bonding between the first and second contact electrodes **57a** and **57b** and the first and second pad electrodes **44a** and **44b**. The anisotropic conductive layer **47** may be formed by attaching an anisotropic conductive adhesive on a protection film, and have a structure in which conductive balls **47a** are diffused in an adhesive resin. Each conductive ball **47a** may be a conductive sphere surrounded by a thin insulation film, and electrically connect a conductor with another conductor as a result of breaking of the insulation film by pressure.

(64) The anisotropic conductive layer **47** may include an anisotropic conductive film (ACF) being in a film type and an anisotropic conductive paste (ACP) being in a paste type.

(65) According to an embodiment of the disclosure, the anisotropic conductive layer **47** may be provided as an anisotropic conductive film.

(66) Accordingly, the insulating films of the conductive balls **47a** may be broken by pressure applied to the anisotropic conductive layer **47** upon mounting of the plurality of inorganic light-emitting diodes **50** onto the substrate **40**, and thus, the first and second contact electrodes **57a** and **57b** of the inorganic light-emitting diodes **50** may be electrically connected with the first and second pad electrodes **44a** and **44b** of the substrate **40**.

(67) However, the plurality of inorganic light-emitting diodes **50** may be mounted on the substrate **40** through a solder, instead of the anisotropic conductive layer **47**. After the inorganic light-emitting diodes **50** are arranged on the substrate **40**, a reflow process may be performed to bond the inorganic light-emitting diodes **50** with the substrate **40**.

(68) The plurality of inorganic light-emitting diodes **50** may include a red light-emitting device **51**, a green light-emitting device **52**, and a blue light-emitting device **53**, and the inorganic light-emitting diodes **50** may be mounted on the mounting surface **41** of the substrate **40** by grouping a series of a red light-emitting device **51**, a green light-emitting device **52**, and a blue light-emitting device **53** into one unit. The series of the red light-emitting device **51**, the green light-emitting device **52**, and the blue light-emitting device **53** may form one pixel. In this case, the red light-emitting device **51**, the green light-emitting device **52**, and the blue light-emitting device **53** may form sub pixels respectively.

(69) The red light-emitting device **51**, the green light-emitting device **52**, and the blue light-emitting device **53** may be arranged at preset intervals in a line, as in the current embodiment of the disclosure, or may be arranged in another type such as a triangle type.

(70) The substrate **40** may include a light absorbing layer **44c** for absorbing external light to improve contrast. The light absorbing layer **44c** may be formed on the entire mounting surface **41** of the substrate **40**. The light absorbing layer **44c** may be formed between the TFT layer **44** and the anisotropic conductive layer **47**.

(71) The plurality of display modules **30A** to **30P** may further include the black matrix **48** formed between the plurality of inorganic light-emitting diodes **50**.

(72) The black matrix **48** may function to complement the light absorbing layer **44c** formed on the entire mounting surface **41** of the substrate **40**. That is, the black matrix **48** may absorb external light such that the substrate **40** is shown to be black, thereby improving contrast of a screen.

(73) The black matrix **48** may have a black color.

(74) In the current embodiment of the disclosure, the black matrix **48** may be positioned between pixels each formed by a series of a red light-emitting device **51**, a green light-emitting device **52**, and a blue light-emitting device **53**. However, unlike the current embodiment of the disclosure, the black matrix **48** may be more delicately formed in such a way as to partition the red light-emitting device **51**, the green light-emitting device **52**, and the blue light-emitting device **53** that are sub pixels.

(75) The black matrix **48** may be formed in a lattice type having a horizontal pattern and a vertical pattern to be positioned between the pixels.

(76) The black matrix **48** may be formed by applying a light absorbing ink onto the anisotropic conductive layer **47** through an ink-jet process and then hardening the light absorbing ink, or the black matrix **48** may be formed by coating the anisotropic conductive layer **47** with a light absorbing film.

(77) That is, the black matrix **48** may be formed at an area between the plurality of inorganic light-emitting diodes **50** on the anisotropic conductive layer **47** formed on the entire mounting surface **41**, wherein none of the plurality of inorganic light-emitting diodes **50** is mounted at the area.

(78) Each of the plurality of display modules **30A** to **30P** may include a front cover **70** positioned on the mounting surface **41** in the first direction X to cover the mounting surface **41** of the display module **30A** to **30P**.

(79) A plurality of front covers **70** may be formed respectively on the plurality of display modules **30A** to **30P** in the first direction X (see FIGS. 6 and 7).

(80) After the individual front covers **70** are formed, the plurality of display modules **30A** to **30P** may be assembled. That is, in examples of the first display module **30A** and the second display module **30B** among the plurality of display modules **30A** to **30P**, a first front cover **70A** may be formed on the mounting surface **41** of the first display module **30A**, and a second front cover **70B** may be formed on the mounting surface **41** of the second display module **30B**.

(81) The front cover **70** may cover the substrate **40** to protect the substrate **40** from an external force or external moisture.

(82) A plurality of layers of the front cover **70** may be provided as a functional film having optical performance. Details about this will be described below.

(83) A part of the plurality of layers of the front cover **70** may include a base layer formed with an Optical Clear Resin (OCR). The base layer may support the plurality of other layers. The OCR may be in a very transparent state having transmittance of 90% or more.

(84) The OCR may improve visibility and image quality by raising transmittance through low reflective properties. That is, in a structure having an air gap, light loss may occur by a difference between reflective indexes of a film layer and an air layer, whereas in a structure using an OCR, light loss may be reduced by a small difference between reflective indexes, resulting in an improvement of visibility and image quality.

(85) That is, the OCR may protect the substrate **40** while improving image quality.

(86) In the front cover **70**, a part of the plurality of layers may include an adhesive layer for adhering the front cover **70** to the mounting surface **41** of the substrate **40**.

(87) The front cover **70** may have a preset height or more in the first direction X which the mounting surface **41** or the light-emitting surface **54** faces.

(88) The reason may be to sufficiently fill up a gap that may be formed between the front cover **70** and the plurality of inorganic light-emitting diodes **50** upon forming of the front cover **70** on the substrate **40**.

(89) Each of the plurality of display modules **30A** to **30P** may include a metal plate **60** positioned on the rear surface **43** of the substrate **40**.

(90) Also, each of the plurality of display modules **30A** to **30P** may include a rear adhesive tape **61** positioned between the rear surface **43** and the metal plate **60** to adhere the metal plate **60** to the rear surface **43** of the substrate **40**.

(91) The rear adhesive tape **61** may be a double-sided adhesive tape, although not limited thereto. However, an adhesive layer may be provided instead of a tape. That is, the rear adhesive tape **61** may be an embodiment of a medium for adhering the metal plate **60** to the rear surface **43** of the substrate **40**, and may be one of various mediums without being limited to a tape.

(92) The plurality of inorganic light-emitting diodes **50** may be electrically connected with an upper wiring layer extending from a pixel driving wire formed on the mounting surface **41** through the side surface **45** of the substrate **40**, the upper wiring layer being formed as a pixel driving wire.

(93) The upper wiring layer may be formed below the anisotropic conductive layer **47**. The upper wiring layer may be electrically connected with a side wire **46** formed on the side surface **45** of the substrate **40**. The side wire **46** may be provided in a thin film type. The side wire **46** may include a coating member **46a** surrounding the side wire **46** to prevent the side wire **46** exposed to outside from being damaged (see FIG. 7).

(94) The side wire **46** may extend to the rear surface **43** of the substrate **40** along the chamfer portion **49** and the side surface **45** of the substrate **40** in the third direction Z. That is, the side wire **46** may extend from the upper edge **32** and the lower edge **34** to the rear surface **43** of the substrate **40** along the chamfer portion **49** and the side surface **45** of the substrate **40**.

(95) The upper wiring layer may be connected with the side wire **46** by an upper connecting pad formed at each edge E of the substrate **40**.

(96) The side wire **46** may extend along the side surface **45** of the substrate **40** and be connected with a rear wiring layer **43b** formed on the rear surface **43**.

(97) On the rear wiring layer **43b**, an insulating layer **43c** may be formed in a direction which the rear surface **43** of the substrate **40** faces to cover the rear wiring layer **43b**.

(98) That is, the plurality of inorganic light-emitting diodes **50** may be electrically connected with the upper wiring layer, the side wire **46**, and the rear wiring layer **43b** sequentially.

(99) Also, as shown in FIG. 4, the display module **30A** may include a driving circuit board **80** for electrically controlling the plurality of inorganic light-emitting diodes **50** mounted on the mounting surface **41**. The driving circuit board **80** may be a printed circuit board. The driving circuit board **80** may be positioned on the rear surface **43** of the substrate **40** in the first direction X. The driving circuit board **80** may be positioned on the metal plate **60** adhered on the rear surface **43** of the

substrate **40**.

(100) The display module **30A** may include a flexible film **81** connecting the driving circuit board **80** with the rear wiring layer **43b** to electrically connect the driving circuit board **80** with the plurality of inorganic light-emitting diodes **50**.

(101) More specifically, one end of the flexible film **81** may be connected with a rear connecting pad **43d** positioned on the rear surface **43** of the substrate **40** and electrically connected with the plurality of inorganic light-emitting diodes **50**.

(102) The rear connecting pad **43d** may be electrically connected with the rear wiring layer **43d**. Accordingly, the rear connecting pad **43d** may electrically connect the rear wiring layer **43b** with the flexible film **81**.

(103) The flexible film **81** may be electrically connected with the rear connecting pad **43d** to transfer power and electrical signals from the driving circuit board **80** to the plurality of inorganic light-emitting diodes **50**.

(104) The flexible film **81** may be a Flexible Flat Cable (FFC), a Chip On Film (COF), etc.

(105) The flexible film **81** may include a first flexible film **81a** and a second flexible film **81b** positioned in the upper and lower directions with respect to the first direction X which is the front direction.

(106) The first and second flexible films **81a** and **81b** may be positioned in the left and right directions or in at least two directions of the upper, lower, left, and right directions with respect to the first direction X, although not limited thereto.

(107) A plurality of second flexible films **81b** may be provided, although not limited thereto. However, the second flexible film **81b** may be provided as a single piece, and also, a plurality of first flexible films **81a** may be provided.

(108) The first flexible film **81a** may transfer a data signal from the driving circuit board **80** to the substrate **40**. The first flexible film **81a** may be a COF.

(109) The second flexible film **81b** may transfer power from the driving circuit board **80** to the substrate **40**. The second flexible film **81b** may be a FFC.

(110) However, the first and second flexible films **81a** and **81b** may be formed in reverse.

(111) The driving circuit board **80** may be electrically connected with the main board **25** (see FIG. 2). The main board **25** may be positioned behind the frame **15**, and the main board **25** may be connected with the driving circuit board **80** through a cable behind the frame **15**.

(112) On a rear surface of the metal plate **60**, a fixing member **82** for adhering the display modules **30A** to **30P** to the frame **15** may be positioned. The fixing member **82** may be a double-sided tape. The metal plate **60** forming a rear side of the display modules **30A** to **30P** may be adhered directly to the frame **15** by the fixing member **82** such that the display modules **30A** to **30P** are supported by the frame **15**.

(113) As described above, the metal plate **60** may be in contact with the substrate **40**. The metal plate **60** may be adhered to the substrate **40** by the rear adhesive tape **61** positioned between the rear surface **43** of the substrate **40** and the metal plate **60**.

(114) FIG. 5 shows the substrate **40** from which some components such as the anisotropic conductive layer **47** are omitted for convenience of description. Also, the side wire **46** may include the coating member **46a** for protecting the side wire **46** from the outside, and the coating member **46a** is omitted for convenience of description.

(115) The metal plate **60** may be formed of a metal material having high heat conductivity. For example, the metal plate **60** may be formed of an aluminum material.

(116) Heat generated from the TFT layer **44** and the plurality of inorganic light-emitting diodes **50** mounted on the substrate **40** may be transferred to the metal plate **60** through the rear adhesive tape **61** along the rear surface **43** of the substrate **40**.

(117) Accordingly, heat generated from the substrate **40** may be easily transferred to the metal plate **60** to prevent temperature of the substrate **40** from rising to preset temperature or more.

(118) The plurality of display modules **30A** to **30P** may be arranged at various locations in a $M \times N$ matrix form. The individual display modules **30A** to **30P** may be movable independently. In this case, each of the display modules **30A** to **30P** may include the metal plate **60** to maintain a certain level of heat-radiating performance regardless of the locations of the display modules **30A** to **30P**.

(119) The plurality of display modules **30A** to **30P** may be arranged in various $M \times N$ matrix forms to implement various screen sizes of the display apparatus **1**. Accordingly, radiating heat from the individual display modules **30A** to **30P** by including the metal plate **60** in each of the display modules **30A** to **30P** as in an embodiment of the disclosure may more improve total heat-radiating performance of the display apparatus **1** than radiating heat through a single metal plate provided for temporary heat radiation.

(120) In a case in which a single metal plate is positioned inside the display apparatus **1**, the metal plate may not exist at a location corresponding to a location of some display modules in a front-rear direction while existing at a location where no display module is positioned, resulting in deterioration of heat-radiating efficiency of the display apparatus **1**.

(121) That is, all of the individual display modules **30A** to **30P** may radiate heat through the metal plate **60** positioned in each of the display modules **30A** to **30P** regardless of the locations, which leads to an improvement of total heat-radiating performance of the display apparatus **1**.

(122) The metal plate **60** may be provided in a shape of a quadrangle substantially corresponding to the shape of the substrate **40**.

(123) An area of the substrate **40** may be larger than or equal to an area of the metal plate **60**. According to a parallel arrangement in first direction X of the substrate **40** and the metal plate **60**, the four edges E of the substrate **40** being in a shape of a rectangle may correspond to four edges of the metal plate **60** with respect to centers of the substrate **40** and the metal plate **60**, or the four edges E of the substrate **40** may be positioned at outer locations than the four edges of the metal plate **60** with respect to the centers of the substrate **40** and the metal plate **60**.

(124) The four edges E of the substrate **40** may be positioned at the outer locations than the four edges of the metal plate **60**. That is, the area of the substrate **40** may be larger than the area of the metal plate **60**.

(125) Upon transferring of heat to the individual display modules **30A** to **30P**, the substrate **40** and the metal plate **60** may be heat-expanded, wherein a degree of expansion of the metal plate **60** may be greater than a degree of expansion of the substrate **40** because the metal plate **60** has a greater coefficient of thermal expansion than the substrate **40**.

(126) In a case in which the four edges E of the substrate **40** correspond to the four edges or the metal plate **60** or are positioned at inner locations than the four edges of the metal plate **60**, the edges of the metal plate **60** may protrude outward from the substrate **40**.

(127) As a result, lengths of gaps formed between the display modules **30A** to **30P** may become non-uniform by thermal expansion of the metal plate **60** of each of the display modules **30A** to **30P**, and accordingly, recognition of some seams may rise, which deteriorates a sense of unity of a screen of the display panel **20**.

(128) However, in a case in which the four edges E of the substrate **40** are positioned at the outer locations than the four edges of the metal plate **60**, the metal plate **60** may not protrude outward from the four edges E of the substrate **40** although the substrate **40** and the metal plate **60** are heat-expanded, and accordingly, the lengths of the gaps formed between the display modules **30A** to **30P** may be maintained uniform.

(129) According to an embodiment of the disclosure, the area of the substrate **40** may substantially correspond to the area of the metal plate **60**. Accordingly, heat generated from the substrate **40** may be radiated uniformly throughout the entire area of the substrate **40** without being isolated at some areas.

(130) The metal plate **60** may be adhered to the rear surface **43** of the substrate **40** by the rear adhesive tape **61**.

(131) The rear adhesive tape **61** may have a size corresponding to the metal plate **60**. That is, an area of the rear adhesive tape **61** may correspond to the area of the metal plate **60**. The metal plate **60** may be substantially in a shape of a quadrangle, and the rear adhesive tape **61** may be in a shape of a quadrangle correspondingly.

(132) The edges of the metal plate **60** being in a shape of a rectangle may correspond to edges of the rear adhesive tape **61** with respect to centers of the metal plate **60** and the rear adhesive tape **61**.

(133) Accordingly, the metal plate **60** and the rear adhesive tape **61** may be easily manufactured as a coupling component, which increases total manufacturing efficiency of the display apparatus **1**.

(134) That is, upon cutting of the metal plate **60** into a plurality of units from a plate, the rear adhesive tape **61** may be first adhered to the metal plate **60** before the metal plate **60** is cut, and then, the rear adhesive tape **61** and the metal plate **60** may be cut together into the plurality of units, thereby reducing a number of processes.

(135) Heat generated from the substrate **40** may be transferred to the metal plate **60** through the rear adhesive tape **61**. Accordingly, the rear adhesive tape **61** may adhere the metal plate **60** to the substrate **40**, while transferring heat generated from the substrate **40** to the metal plate **60**.

(136) Accordingly, the rear adhesive tape **61** may include a material having high heat-radiating performance.

(137) Basically, the rear adhesive tape **61** may include an adhesive material to adhere the metal plate **60** to the substrate **40**.

(138) In addition, the rear adhesive tape **61** may include a material having higher heat-radiating performance than general adhesive materials. Accordingly, the rear adhesive tape **61** may efficiently transfer heat between the substrate **40** and the metal plate **60**.

(139) Also, the adhesive material of the rear adhesive tape **61** may be formed of a material having higher heat-radiating performance than adhesive materials configuring normal adhesives.

(140) The material having the higher heat-radiating material may be a material having high heat conductivity, high heat transfer performance, and low specific heat to effectively transfer heat.

(141) For example, the rear adhesive tape **61** may include a graphite material, although not limited thereto. However, the rear adhesive tape **61** may be any material having high heat-radiating performance.

(142) Flexibility of the rear adhesive tape **61** may be greater than flexibility of the substrate **40** and the metal plate **60**. Accordingly, the rear adhesive tape **61** may be a material having adhesiveness, heat-radiating performance, and high flexibility. The rear adhesive tape **61** may be an inorganic material double-sided tape. Because the rear adhesive tape **61** is an inorganic material double-sided tape, as described above, the rear adhesive tape **61** may be formed as a single layer without any material for supporting one surface that is adhered to the substrate **40** and the other surface that is adhered to the metal plate **60**.

(143) Because the rear adhesive tape **61** includes no material that interferes with heat transfer, heat-radiating performance may be improved. However, the rear adhesive tape **61** is not limited to an inorganic material double-sided tape, and may be a heat-radiating tape having higher heat-radiating performance than normal double-sided tapes.

(144) The rear adhesive tape **61** may be formed of a material having high flexibility to absorb an external force transferred from the substrate **40** and the metal plate **60**. More specifically, the flexibility of the rear adhesive tape **61** may be greater than flexibility of the substrate **40** and the metal plate **60**.

(145) Accordingly, upon transferring of an external force generated by a change in size of the substrate **40** and the metal plate **60**, caused by heat transferred to the substrate **40** and the metal plate **60**, to the rear adhesive tape **61**, the rear adhesive tape **61** may itself be deformed to thereby prevent the external force from being transferred to other components.

(146) The rear adhesive tape **61** may have a preset thickness in the first direction X. Upon expanding or contracting of the metal plate **60** by heat transferred to the metal plate **60**, the metal

plate **60** may be expanded or contracted in a direction that is orthogonal to the first direction X, as well as in the first direction X, and accordingly, an external force may be transferred to the substrate **40**.

(147) Because the metal plate **60** is formed with a size corresponding to the substrate **40** to cover the entire rear surface **43** of the substrate **40**, as described above, the fixing member **82** may be positioned on the rear surface of the metal plate **60**, although not limited thereto.

(148) However, the fixing member **82** may be positioned on the rear surface **43** of the substrate **40**. In this case, the substrate **40** may be adhered directly to the frame **15** through the fixing member **82**.

(149) Unlike an embodiment of the disclosure, the metal plate **60** may cover a portion of the rear surface **43** of the substrate **40**, and the fixing member **82** may be adhered to an area at which the metal plate **60** does not cover the rear surface **43** of the substrate **40**.

(150) The fixing member **82** may be a double-sided tape.

(151) Hereinafter, the front cover **70**, a side cover **90**, and a side member **100** will be described in detail.

(152) FIG. **6** is a cross-sectional view showing some components of the display apparatus of FIG. **1** in a third direction, FIG. **7** is an enlarged cross-sectional view of the some components shown in FIG. **6**, and FIG. **8** schematically shows a front side of some components of the display apparatus of FIG. **1**.

(153) The front cover **70** may protect the substrate **40** from an external force, deteriorate recognition of seams formed by gaps G between the plurality of display modules **30A** to **30P**, and improve color deviation between the plurality of display modules **30A** to **30P**.

(154) Each of the plurality of display modules **30A** to **30P** may include the side cover **90** positioned in a gap G formed between the plurality of display modules **30A** to **30P** upon an arrangement of the plurality of display modules **30A** to **30P**.

(155) To absorb light reflected from the gaps G between the plurality of display modules **30A** to **30P**, the front cover **70** of each of the plurality of display modules **30A** to **30P** may extend outward from the substrate **40** of each of the plurality of display modules **30A** to **30P**. A side **75** of the front cover **70** may extend to an outer location than the mounting surface **41**.

(156) More specifically, the front cover **70** may extend to an outer location than an edge (side) **41S** of the mounting surface **41** of the substrate **40** in the second direction Y and the third direction Z.

(157) Substantially, a gap between the display modules **30A** to **30P** may be formed between the side surfaces **45** of the substrates **40** of the display modules **30A** to **30P**, however, according to an embodiment of the disclosure, a gap G may be a non-display area that may be formed between the display modules **30A** to **30P**. Therefore, the gap G formed between the plurality of display modules **30A** to **30P** may be interpreted to be a spacing formed between an edge **41S** of a mounting surface **41** of a substrate **40** of a display module **30A** to **30P** and an edge **41S** of a mounting surface **41** of a substrate **40** of another display module **30A** to **30P** neighboring the corresponding display module **30A** to **30P**.

(158) Accordingly, the gap G formed between the plurality of display modules **30A** to **30P** may be a spacing formed between the edge **41S** of the mounting surface **41** of the display module **30A** to **30P** and the edge **41S** of the mounting surface **41** of the other display module **30A** to **30P** neighboring the corresponding display module **30A** to **30P** in the second direction Y or the third direction Z.

(159) In the gap G between the plurality of display modules **30A** to **30P**, the front cover **70** extending from each of the display modules **30A** to **30P** may be positioned to absorb light radiated to the gap G or light reflected in the gap G, thereby reducing recognition of seams.

(160) Also, the side cover **90** of each of the plurality of display modules **30A** to **30P**, the side cover **90** positioned in the gap G, may absorb light radiated to the gap G, thereby reducing visibility of seams, which will be described below.

(161) As shown in FIGS. 6 and 7, the front cover **70** may extend to an outer location than the substrate **40** in the third direction Z. More specifically, the front cover **70** may extend to an outer location than the side surface **45** and the chamfer portion **49** in the third direction Z.

(162) According to an embodiment of the disclosure, one edge of the substrate **40**, corresponding to the lower edge **34** of the first display module **30A** will be described, however, the front cover **70** may extend to an outer location than the four edges E of the substrate **40** in the second direction Y or the third direction Z.

(163) That is, the side **75** of the front cover **70**, corresponding to an edge of the front cover **70**, may extend to an outer location of the substrate **40**, more specifically, the mounting surface **41** than the four edges E of the substrate **40** in the second direction Y or the third direction Z.

(164) The front cover **70** may include a plurality of layers having different optical properties. The plurality of layers may have a structure resulting from stacking layers in the first direction X.

(165) The plurality of layers may be bonded in the first direction X to constitute the front cover **70**.

(166) A layer among the plurality of layers may be an anti-glare layer, although not limited thereto. However, the layer may be an anti-reflective layer or a combination of an anti-glare layer and an anti-reflective layer.

(167) Another layer among the plurality of layers may be a transmission rate adjusting layer, although not limited thereto. However, the other layer may be a layer having another physical property, another material, or another function. For example, the other layer may be a circularly polarized layer.

(168) Alternatively, in another embodiment, a single layer may be used instead of the plurality of layers. The single layer may be a layer capable of functionally implementing all functions of the plurality of layers.

(169) The front cover **70** may include an adhesive layer as described above. The adhesive layer may be positioned at a lowermost location of the plurality of layers in the first direction X and adhered to the mounting surface **41**. The adhesive layer may have a preset height or more in the first direction X which the mounting surface **41** or the light-emitting surface **54** faces to sufficiently fill up a gap that may be formed between the adhesive layer and the inorganic light-emitting diodes **50** upon adhering of the adhesive layer to the substrate **40**.

(170) The adhesive layer is not limited to the embodiment of the disclosure, and the adhesive layer may be provided as a separate component from the front cover **70** and positioned between the front cover **70** and the mounting surface **41** to adhere the front cover **70** to the mounting surface **41**.

(171) Accordingly, because the front cover **70** is closely adhered to the mounting surface **41** to protect components mounted on the mounting surface **41**, the display module **30** may adhere the front cover **70** directly to the substrate **40** without an additional molding component formed between the front cover **70** and the substrate **40**.

(172) The front cover **70** may diffuse and reflect incident light from the outside to prevent the incident light from being specularly reflected to a user's eyes.

(173) By diffusing and reflecting incident light from the outside, a glaring effect may deteriorate, and accordingly, contrast of a screen displayed on the display panel **20** may be improved.

(174) Also, the front cover **70** may deteriorate transmittance of incident external light or external light reflected from the substrate **40** and the gap G.

(175) The front cover **70** according to an embodiment of the disclosure may include a material that deteriorates transmittance of light to absorb at least a part of light transmitted toward the substrate **40** or reflected light reflected from the substrate **40** and traveling in the first direction X.

(176) Upon manufacturing of a plurality of substrates, some substrates may have different colors due to errors in a manufacturing process. Accordingly, substrates having different unique colors may be tiled to constitute a single display panel.

(177) As described above, the front cover **70** according to an embodiment of the disclosure may absorb at least a part of light reflected from the substrate **40** and transmitted to the outside, thereby

raising a sense of unity of a screen of the display panel **20**.

(178) That is, the front cover **70** may deteriorate color deviation of the plurality of display modules **30A** to **30P**, generated in a process for the display modules **30A** to **30P**, by deteriorating transmittance with respect to external light.

(179) The front cover **70** may improve contrast of a screen displayed on the display panel **20** by preventing external light entered the display panel **20** from the outside from being transmitted to the substrate **40**, and additionally absorbing a part of light entered the display panel **20** from the outside or absorbing a part of external light reflected from the substrate **40** and transmitted to the outside of the display panel **20**. Such different optical actions may be implemented through the plurality of layers described above.

(180) That is, the front cover **70** may be positioned in front of the substrate **40** in the first direction **X** to improve contrast that may deteriorate by external light in a screen displayed on the display panel **20**.

(181) In the display module **30** according to an embodiment of the disclosure, the front cover **70** may extend to the outer location from the substrate **40** in the third direction **Z**, as described above.

(182) Accordingly, a part of light entered the gap **G** formed between the plurality of display modules **30A** to **30P** may be blocked by at least one portion of the front cover **70** positioned above the gap **G**, and at least a part of external light entered the gap **G** or external light reflected in the gap **G** may be absorbed by the front cover **70** positioned above the gap **G** and thus prevented from being transmitted to the outside. Accordingly, recognition of a seam formed in the gap **G** may deteriorate, and according to the deterioration of recognition of the seam, a sense of unity of a screen displayed on the display panel **20** may be improved.

(183) More specifically, the side **75** of the front cover **70** in the third direction **Z** may be positioned at an outer location than the edge **41S** of the mounting surface **41** in the second direction **Z**, or above the gap **G**.

(184) Accordingly, the front cover **70** may include a first area **71** positioned at the outer location than the edge **41S** of the mounting surface **41** in the third direction **Z**, or above the gap **G**, and a second area **72** positioned on the mounting surface **41**.

(185) The first area **71** and the second area **72** of the front cover **70** may be divided by the gap **G** in the third direction **Z**.

(186) The first area **71** of the front cover **70** may be positioned above the gap **G** to block external light radiated toward the gap **G** or prevent light reflected in the gap **G** from being radiated to the outside, thereby reducing recognition of a seam that is a boundary of the plurality of display modules **30A** to **30P** and may be formed by the gap **G**, which leads to an improvement of a sense of unity of the display panel **20**.

(187) The front cover **70** may extend to the outer locations than the four edges **41S** of the mounting surface **41** of the substrate **40**, as described above, resulting in a deterioration of recognition of seams that may be formed at the edges of the plurality of display modules **30A** to **30P**.

(188) For example, in the first display module **30A** and the second display module **30B**, a first area **71A** of the first front cover **70A** of the first display module **30A**, the first area **71A** extending from the first display module **30A**, may be positioned in a gap **G** formed between the first display module **30A** and the second display module **30B**.

(189) Above the gap **G**, a side **75A** of the first front cover **70A** of the first display module **30A** and a side **75B** of a second front cover **70B** of the second display module **30B** neighboring the first display module **30A** may be positioned.

(190) Also, in the gap **G**, the side surface **45** and the chamfer portion **49** of each of the first and second display modules **30A** and **30B** may be positioned.

(191) A second area **72A** of the first front cover **70A** of the first display module **30A** may be positioned on the mounting surface **41** of the first display module **30A**.

(192) A first area **71B** of the second front cover **70B** of the second display module **30B**, the first

area **71B** extending from the second display module **30B**, may be positioned above the gap **G** formed between the first display module **30A** and the second display module **30B**, and a second area **72B** of the second front cover **70B** of the second display module **30B** may be positioned on the mounting surface **41** of the second display module **30B**.

(193) That is, above the gap **G** formed between the first display module **30A** and the second display module **30B**, the first area **71A** of the first front cover **70A** of the first display module **30A** and the first area **71B** of the second front cover **70B** of the second display module **30B** may be aligned in the third direction **Z**.

(194) A length of each of the first areas **71A** and **71B** of the first and second front covers **70A** and **70B** of the first and second display modules **30A** and **30B**, extending in the third direction **Z**, may be about half or less of a length of the gap **G**.

(195) Accordingly, upon an alignment in third direction **Z** of the first areas **71A** and **71B** of the first and second front covers **70A** and **70B** of the first and second display modules **30A** and **30B**, a sum of lengths of the first areas **71A** and **71B** may be substantially equal to or smaller than the length of the gap **G**.

(196) According to an embodiment of the disclosure, upon the alignment in third direction **Z** of the first areas **71A** and **71B** of the first and second front covers **70A** and **70B** of the first and second display modules **30A** and **30B**, a certain spacing may be formed between the side **75A** of the first front cover **70A** of the first display module **30A** and the side **75B** of the second front cover **70B** of the second display module **30B**. The spacing may be made by the side member **100** positioned at the side of each of the display modules **30A** to **30P**. The side member **100** will be described below.

(197) As described above, the first area **71A** of the first front cover **40A** of the first display module **30A** and the first area **71B** of the second front cover **70B** of the second display module **30B** may be positioned above the gap **G** between the first display module **30A** and the second display module **30B**.

(198) External light entering the display panel **20** may be transmitted through the first areas **71A** and **71B** of the first and second front covers **70A** and **70B** of the first and second display modules **30A** and **30B** and diffused and reflected to the outside of the display panel **20**, or a part of the external light may be absorbed in the first areas **71A** and **71B**. Accordingly, an amount of light arrived at the gap **G** may be reduced, and recognition of a boundary between the first display module **30A** and the second display module **30B** by the gap **G** may be reduced.

(199) Also, light reflected in the gap **G** and traveling toward the outside of the display panel **20** may be transmitted through the first areas **71A** and **71B** of the first and second front covers **70A** and **70B** of the first and second display modules **30A** and **30B** and diffused and reflected to the outside of the display panel **20**, or a part of the light may be absorbed in the first areas **71A** and **71B**. Accordingly, an amount of light transmitted to the outside of the display panel **20** may be reduced, and recognition of a boundary between the first display module **30A** and the second display module **30B** by the gap **G** may be reduced.

(200) That is, by reducing an amount of external light entering the gap **G** formed between the plurality of display modules **30A** to **30P** while absorbing at least a part of external light reflected in the gap **G**, a sense of unity of the display panel **20** may be improved.

(201) Additionally, because at least a part of external light reflected from the substrates **40A** and **40B** to be displayed to the outside is absorbed in the first and second front covers **70A** and **70B** of the first and second display modules **30A** and **30B** such that unique colors of the substrates **40A** and **40B** are not recognized from the outside although the substrate **40A** of the first display module **30A** and the substrate **40B** of the second display module **30B** have different colors, a sense of unity of the display panel **20** may be improved.

(202) The first display module **30A** may include the side cover **90** positioned below the front cover **70** in the direction in which the mounting surface **41** faces and formed on the side surface **45** of the substrate **40**.

(203) More specifically, the side cover **90** may be positioned in a space defined by a lower surface **76** of the first area **71** of the front cover **70** in the first direction X, a lower surface of the anisotropic conductive layer **47**, and a side surface of the substrate **40** in the third direction Z.

(204) Also, a side **47S** of the anisotropic conductive layer **47** of the display module **30** may be aligned with the side **75** of the front cover **70** in the first direction X. In this case, the side cover **90** may be positioned in a space defined by the lower surface of the anisotropic conductive layer **47** in the first direction X and the side surface of the substrate **40** in the third direction Z.

(205) Also, the side **47S** of the anisotropic conductive layer **47** of the display module **30** may be aligned with the edge **41S** of the mounting surface **41** in the first direction X. In this case, the side cover **90** may be positioned in a space defined by the lower surface **76** of the first area **71** of the front cover **70** in the first direction X and the side surface of the substrate **40** in the third direction Z.

(206) The side cover **90** may be adhered to the lower surface **76** of the first area **71**, the side surface **45**, and at least one portion of the metal plate **60**. The side cover **90** may be adhered to the entire lower surface **76** of the first area **71**. Also, the side cover **90** may cover the entire area of the side surface **45**.

(207) The lower surface **76** of the first area **71** may be at least one area of an entire lower surface of the front cover **70**, and may be a rear surface of the adhesive layer formed at the lowermost location of the front cover **70**.

(208) Also, the side cover **90** may cover a pair of chamfer portions **49** positioned in a front-rear direction of the side surface **45** in the first direction X.

(209) The side cover **90** may surround the entire of the chamfer portion **49** formed between the mounting surface **41** and the side surface **45**, as well as the side surface **45**.

(210) Because the side cover **90** surrounds the chamfer portion **49** formed between the mounting surface **41** and the side surface **45**, the side cover **90** may fill up a space that may be formed between the substrate **40** and the front cover **70**.

(211) Accordingly, the side cover **90** may prevent a foreign material or water from entering the space between the substrate **40** and the front cover **70** from the outside.

(212) Also, because the side cover **90** surrounds the chamfer portion **49** formed between the rear surface **43** and the side surface **45**, the side cover **90** may fill up a space that may be formed between the substrate **40** and the metal plate **60**.

(213) Accordingly, the side cover **90** may prevent a foreign material or water from entering the space between the substrate **40** and the metal plate **60** from the outside.

(214) The side cover **90** may be in contact with the lower surface **76** of the first area **71**, the chamfer portion **49** of the substrate **40**, and the side surface **45**. Accordingly, the side cover **90** may support the lower surface **76** of the first area **71**, the chamfer portion **49** of the substrate **40**, and the side surface **45**.

(215) The front cover **70** may be adhered to the substrate **40** by the front cover **70**, as described above, and the side cover **90** may reinforce adhesiveness between the front cover **70** and the substrate **40**. Accordingly, the side cover **90** may prevent the front cover **70** from separating from the substrate **40**.

(216) That is, reliability of the first display module **30A** may be raised by the side cover **90**.

(217) Also, the metal plate **60** may be adhered to the substrate **40** by the rear adhesive tape **61**, and the side cover **90** may reinforce adhesiveness between the metal plate **60** and the substrate **40**. Accordingly, the side cover **90** may prevent the metal plate **60** from separating from the substrate **40**.

(218) The side surface **45** of the substrate **40** may correspond to the four edges **41S** of the mounting surface **41**, and the first area **71** of the front cover **70** may extend to the outer locations than the four edges **41S** of the mounting surface **41** in the second direction Y and the third direction Z in which the mounting surface **41** extends.

(219) The side cover **90** may surround the lower surface **76** of the first area **71** and the side surface **46** corresponding to each of the four edges **41S** of the mounting surface **41** along a circumference of the four edges **41S** of the mounting surface **41**.

(220) That is, the side cover **90** may seal the entire edges of portions at which the substrate **40** is adhered to the front cover **70**.

(221) The side cover **90** may cover the lower surface **76** of the first area **71** and the side surface **45** in all directions that are orthogonal to the first direction X.

(222) Accordingly, coupling between the front cover **70** and the substrate **40** may be improved, and the front cover **70** and the side surface **45** of the substrate **40** may be protected from an external force.

(223) Also, the side cover **90** may prevent external water or a foreign material from entering between the substrate **40** and the front cover **70**, as described above. In addition, the side cover **90** may prevent external water or a foreign material from entering a gap formed between the substrate **40** and the front cover **70** due to low adhesiveness.

(224) The side cover **90** may surround the four edges E of the substrate **40** along the side surface **45** of the substrate **40** to seal between the substrate **40**, the front cover **70**, and the metal plate **60**.

(225) Accordingly, the side cover **90** may prevent a foreign material or water entered in all directions from entering the substrate **40** and the front cover **70**.

(226) As described above, because the lowermost end of the front cover **70** in the first direction X is provided as an adhesive layer, the lower surface **76** of the first area **71** may be provided as a rear surface of the adhesive layer.

(227) Accordingly, upon exposure of the lower surface **76** of the first area **71** to the outside, a foreign material existing in the outside may be adhered to the lower surface **76** of the first area **71**.

(228) Upon an arrangement of the plurality of display modules **30A** to **30P** in a state in which a foreign material is adhered to the lower surface **76** of the first area **71**, recognition of a seam generated between the plurality of display modules **30A** to **30P** may increase due to the foreign material adhered to the lower surface **76** of the first area **71**.

(229) However, because the display module **30A** according to an embodiment of the disclosure includes the side cover **90** and the side cover **90** covers the lower surface **76** of the first area **71**, it may be possible to prevent a foreign material from being adhered to the lower surface **76** of the first area **71**.

(230) Accordingly, it may be possible to reduce visibility of a seam generated between the plurality of display modules **30A** to **30P**, caused by a foreign material adhered to the front cover **70** upon an arrangement of the plurality of display modules **30A** to **30P**.

(231) Also, current may enter a plurality of electronic components mounted on the substrate **40** by an electrostatic discharge that may occur on the display modules **30A** to **30P** to damage the electronic components, which will be described below. The side cover **90** may prevent charges generated by an electrostatic discharge from entering the substrate **40** by sealing the substrate **40** from the outside to prevent the electronic components from being damaged.

(232) That is, because the substrate **40** is sealed by the front cover **70** and the side cover **90** to prevent charges generated by an electrostatic discharge from passing through the front cover **70** and the side cover **90**, the charges may be prevented from flowing to the substrate **40** and charges flowing on the front cover **70** and the side cover **90** may be guided to the metal plate **60** being in contact with the side cover **90**, thereby providing a path of current by an electrostatic discharge. Accordingly, ESD pressure resistance of the electronic components mounted on the substrate **40** may be improved.

(233) The first display module **30A** may be positioned below the front cover **70** in the direction in which the mounting surface **41** faces, as described above. That is, the side cover **90** may be not positioned above the lower surface **76** in the first direction X.

(234) A foremost surface of the side cover **90** in the first direction X may be in contact with the

lower surface **76** of the first area **71**, and may be not positioned in front of the lower surface **76** of the first area **71** in the first direction **X**.

(235) The reason may be not to position the side cover **90** on a traveling path of light radiated from the plurality of inorganic light-emitting diodes **50**.

(236) In a case in which at least one portion of the side cover **90** is positioned in front of the lower surface **76** or in front of the front cover **70** in the first direction **X**, the side cover **90** may be positioned on a path of light traveling in the front direction through the front cover **70**.

(237) That is, the side cover **90** may absorb or diffuse and reflect a part of traveling light to distort some area of an image displayed in the display module **20**.

(238) However, because the side cover **90** according to an embodiment of the disclosure is positioned behind the front cover **70** in the first direction **X**, the side cover **90** may not limit a movement of light radiated by the plurality of inorganic light-emitting diodes **50**, thereby improving image quality of the display panel **20**.

(239) The side **75** of the front cover **70** in the third direction **Z** and the side **91** of the side cover **90** in the third direction **Z** may be substantially aligned in the first direction **X**.

(240) The reason may be because the front cover **70** and the side cover **90** are cut simultaneously in a process of manufacturing the display module **30A**. Also, the side member **100** may be adhered to the side **75** of the front cover **70** and the side **91** of the side cover **90** substantially aligned in the first direction **X**.

(241) That is, a spacing that is formed between the plurality of display modules **30A** to **30P** upon an arrangement of the plurality of display modules **30A** to **30P** may be reduced, and a seam that may be recognized by the spacing between the plurality of display modules **30A** to **30P** may be reduced.

(242) The side cover **90** may include a material that absorbs light. For example, the side cover **90** may be formed of an opaque or translucent material.

(243) Also, the side cover **90** may include a photosensitive material. For example, the side cover **90** may be formed of a photosensitive OCR. According to irradiating of external light such as ultraviolet light (UV) having another wavelength except for a visible light wavelength to the photosensitive material, the photosensitive material may change a physical property to show a dark color.

(244) Accordingly, the side cover **90** may be formed of a material that is colored with a dark color upon irradiating the side cover **90** with ultraviolet light (UV) to during a manufacturing process to absorb light.

(245) The side cover **90** may have a dark color. The side cover **90** may have a color that is darker than the front cover **70**.

(246) The side cover **90** may have a color that is similar to that of the black matrix **48**.

(247) Accordingly, light that enters the side cover **90** may be absorbed in the side cover **90** without being reflected, by the material absorbing light.

(248) The side cover **90** may be positioned in the gap **G** formed between the plurality of display modules **30A** to **30P** together with the first area **71** of the front cover **70** upon an arrangement of the plurality of display modules **30A** to **30P**.

(249) Accordingly, the side cover **90** may absorb light that enters the gap **G** to reduce light that enters the gap **G**, is reflected, and then emitted to the outside. Accordingly, recognition of a seam that is formed by the gap **G** formed between the plurality of display modules **30A** to **30P** may deteriorate.

(250) For example, in the first display module **30A** and the second display module **30B**, a side cover **90A** of the first display module **30A** and a side cover **90B** of the second display module **30B** may be positioned in the gap **G** formed between the first display module **30A** and the second display module **30B** together with the first area **71A** of the first front cover **70A** of the first display module **30A** and the first area **71B** of the second front cover **70B** of the second display module **30B**.

(251) In the gap G, a side **91S** of the side cover **90A** of the first display module **30A** and a side **91B** of the side cover **90B** of the second display module **30B** may be positioned together with the neighboring sides **75A** and **75B** of the first and second front covers **70A** and **70B** of the first and second display modules **30A** and **30B**.

(252) The neighboring sides **75A** and **75B** of the first and second front covers **70A** and **70B** of the first and second display modules **30A** and **30B** and the neighboring sides **91A** and **91B** of the side covers **90A** and **90B** of the first and second display modules **30A** and **30B** may face each other. The neighboring sides **75A** and **75B** of the first and second front covers **70A** and **70B** of the first and second display modules **30A** and **30B** and the neighboring sides **91A** and **91B** of the side covers **90A** and **90B** of the first and second display modules **30A** and **30B** may be positioned in parallel to each other.

(253) That is, in the gap G formed between the first display module **30A** and the second display module **30B**, the first areas **71A** and **71B** of the front covers **70A** and **70B** of the first and second display modules **30A** and **30B** and the side covers **90A** and **90B** of the first and second display modules **30A** and **30B** may be aligned in the third direction Z.

(254) A length of each of the side covers **90A** and **90B** of the first and second display modules **30A** and **30B**, extending in the third direction Z, may be about half or less of the gap G to correspond to the first areas **71A** and **71B** of the front covers **70A** and **70B** of the first and second display modules **30A** and **30B**.

(255) In the gap G between the first display module **30A** and the second display module **30B**, the first area **71A** of the first front cover **70A** of the first display module **30A** and the first area **71B** of the second front cover **70B** of the second display module **30B** may be positioned, and the side covers **90A** and **90B** of the first and second display modules **30A** and **30B** may be positioned behind the first areas **71A** and **71B** in the first direction X.

(256) As described above, external light entered the display panel **20** may be transmitted through the first and second areas **71A** and **71B** of the first and second front covers **70A** and **70B** of the first and second display modules **30A** and **30B** and then diffused and reflected to the outside of the display panel **20**, or a part of the external light may be absorbed in the first areas **71A** and **71B**. As a result, an amount of light arrived at the gap G may be reduced.

(257) In addition, light arrived at the gap G may be absorbed in the side covers **90A** and **90B** of the first and second display modules **30A** and **30B**, positioned in the gap G, and accordingly, recognition of a boundary between the first display module **30A** and the second display module **30B** may be reduced.

(258) That is, by reducing an amount of external light entering the gap G formed between the plurality of display modules **30A** to **30P** while additionally absorbing light arrived at the gap G, a sense of unity of a screen of the display panel **20** may be improved.

(259) In addition, light reflected from the side covers **90A** and **90B** of the first and second display modules **30A** and **30B** and traveling to the outside of the display panel **20** without being absorbed in the side covers **90A** and **90B** may be transmitted through the first areas **71A** and **71B** of the first and second front covers **70A** and **70B** to be diffused and reflected to the outside of the display panel **20**, or a part of the light may be absorbed in the first areas **71A** and **71B**. Accordingly, an amount of light transmitted to the outside of the display panel **20** may be reduced, resulting in a reduction of recognition of a boundary between the first display module **30A** and the second display module **30B** by the gap G.

(260) Because the side cover **90** is arranged in the gap G formed between the plurality of display modules **30A** to **30P** upon an arrangement of the plurality of display modules **30A** to **30P**, as described above, the side cover **90** may absorb light arrived at the gap G to reduce visibility of a seam that may be recognized by the gap G.

(261) According to the above-described example, the front cover **70** may reduce an amount of light arriving at the substrate **40** by diffusing and reflecting, absorbing, or circularly polarizing a part of

light that enters the display module **20** or changing a reflection direction of the light, although embodiments are not limited thereto.

(262) However, the front cover **70** may be formed of a transparent material that transmits light without distorting the light. In this case, the side cover **90** positioned between the plurality of display modules **30A** to **30P** may also reduce recognition of a boundary between the plurality of display modules **30A** to **30P** by the gap **G**.

(263) Because the side cover **90** is formed of a material absorbing light, as described above, a part of light radiated from the plurality of inorganic light-emitting diodes **50** may be absorbed in the side cover **90** in a case in which at least one portion of the side cover **90** is positioned in front of the front cover **70** in the first direction **X**. Accordingly, an area of a screen displayed on the display module **20** may be shown to be dark.

(264) However, because the side cover **90** according to an embodiment of the disclosure is positioned below the front cover **70** in the first direction **X**, more specifically, below the lower surface **76** of the first area **71**, the side cover **90** may not absorb light radiated from the plurality of inorganic light-emitting diodes **50**, and accordingly, brightness of an image displayed on the display module **20** may become uniform.

(265) The anisotropic conductive layer **47** may be an anisotropic conductive film. The anisotropic conductive layer **47** may be bonded with the TFT layer **44** in a film type on the TFT layer **44**.

(266) The anisotropic conductive layer **47** may be formed in a film type, and an area of the anisotropic conductive layer **47** may be larger than an area of the substrate **40**.

(267) Accordingly, after the anisotropic conductive layer **47** is bonded with the TFT layer **44**, a process of cutting the anisotropic conductive layer **47** such that the area of the anisotropic conductive layer **47** corresponds to the area of the substrate **40** may be performed.

(268) The cutting process may be to cut the anisotropic conductive layer **47** to correspond the area of the anisotropic conductive layer **47** to the area of the substrate **40** through laser cutting, etc.

(269) The anisotropic conductive layer **47** may have an area corresponding to an area of the mounting surface **41**. However, because the anisotropic conductive layer **47** is formed as an anisotropic conductive film, as described above, it may be not easy to correspond the area of the anisotropic conductive film to the area of the mounting surface **41**. Also, upon adhering of the anisotropic conductive film corresponding to the area of the mounting surface **41** to the mounting surface **41**, the anisotropic conductive film may have a smaller section than the mounting surface **41** due to manufacturing tolerance, resulting in deterioration of reliability of the display module **30**.

(270) Accordingly, by adhering an anisotropic conductive film having a larger area than the mounting surface **41** to the substrate **40** and then cutting the anisotropic conductive film to an area corresponding to the area of the substrate **40**, the anisotropic conductive layer **47** may be formed.

(271) The side surface **45** of the substrate **40** may be positioned at an outer location than the mounting surface **41** by the chamfer portion **49**. The anisotropic conductive film may be cut based on a side of the coating member **46a** forming a side of the side wire **46** in the third direction **Z**.

(272) The reason may be because there is a risk that the side surface **45**, the chamfer portion **49**, or the side wire **46** of the substrate **40** will be damaged upon cutting of the anisotropic conductive film based on the mounting surface **41**.

(273) However, the anisotropic conductive film may be cut together with the front cover **70** to form the anisotropic conductive layer **47**. In this case, the side **47S** of the anisotropic conductive layer **47** may be aligned with the side **75** of the front cover **70** in the first direction **X**.

(274) In this case, the side **47S** of the anisotropic conductive layer **47** may be exposed to the outside and damaged by static electricity. However, ESD reliability may be secured by the side member **100** which will be described below.

(275) Accordingly, upon cutting of the anisotropic conductive film, the side **47S** of the anisotropic conductive layer **47** may be positioned at an outer location than the mounting surface **41**. More specifically, because the anisotropic conductive film is cut based on the side surface **45** or the side

46S of the side wire **46**, as described above, the side **47S** of the anisotropic conductive layer **47** may be aligned with the side surface **45** or the side **46S** of the side wire **46** in the first direction X. Also, the side **47S** of the anisotropic conductive layer **47** may be positioned at the outer location than the side surface **45** or the side **46S** of the side wire **46** by manufacturing tolerance or burr formed in the anisotropic conductive film upon cutting.

(276) However, to prevent the substrate **40** from being damaged in an actual cutting process, a cutting location of the anisotropic conductive film may be an outer location than the side surface **45** or the side **46S** of the side wire **46**.

(277) Accordingly, the side **47S** of the anisotropic conductive layer **47** may be formed outside the substrate **40**. Particularly, the side **47S** of the anisotropic conductive layer **47** may be positioned at an outer location than the side cover **90**.

(278) The side cover **90** may cover an outer side of the side surface **45** in the second direction Y, as well as an outer side of the side surface **45** of the substrate **40** in the third direction Z, as shown in FIG. 7.

(279) That is, the side cover **90** may surround all the four edges E of the substrate **40**, as described above.

(280) Accordingly, the mounting surface **41** which is the front surface of the substrate **40** may be covered by the front cover **70**, the rear surface **43** of the substrate **40** may be covered by the metal plate **60**, and the side surface **45** and the chamfer portion **49** of the substrate **40** may be covered by the side cover **90**.

(281) The front cover **70** may be formed of a non-conductive material through which no charges are transmitted.

(282) The side cover **90** may be formed of a non-conductive material through which no charges are transmitted.

(283) Because the front cover **70** and the side cover **90** are formed of a non-conductive material, a major part of current applied to the front cover **70** or the side cover **90** may flow on the front cover **70** and the side cover **90** without being transmitted through the front cover **70** and the side cover **90**.

(284) The metal plate **60** may be formed of a material having great capacitance, and function as a ground. Accordingly, upon applying of current to the metal plate **60**, the metal plate **60** may be maintained at a constant potential. The current applied to the metal plate **60** may be absorbed in the metal plate **60**, and no current may flow to the substrate **40** through the metal plate **60**.

(285) Also, the entire side wire **46** of the substrate **40** may be surrounded by the side cover **90**, and accordingly, the side wire **46** may be sealed not to be exposed to the outside. Accordingly, static electricity discharged from the side surface **45** of the substrate **40** may not enter the side wire **46** due to the side cover **90**.

(286) In a display apparatus manufacturing process of implementing a display panel with display modules, the display panel may be formed by tiling a plurality of display modules. During a process for forming the display panel with the display modules, current generated by an electrostatic discharge may enter the inside of the display modules while the display modules are manufactured and conveyed, which may damage electronic components mounted inside the display modules.

(287) Particularly, due to a failure generated during a process of manufacturing a display module, a side wire extending along a side surface of a substrate may be exposed to the outside or a spacing may be made between an anisotropic conductive layer and a front cover or the substrate, and during a process of applying a side cover and hardening the side cover, a spacing may be made inside the side cover. In this case, due to a process failure caused by an electrostatic discharge, current may enter electronic components such as the side wire mounted on the substrate to damage the electronic components.

(288) Each of the display modules **30A** to **30P** may include a component configured to prevent

current generated by an electrostatic discharge from entering components mounted on the substrate **40**, and accordingly, current generated by an electrostatic discharge may be easily guided to the metal plate **60** which is a ground component along the front cover **70** and the side cover **90** sealing the substrate **40** on each of the display modules **30A** to **30P**, without entering the components mounted on the substrate **40**.

(289) The display apparatus **1** according to an embodiment of the disclosure may further include the side member **100** positioned on an outer side of the side cover **90** of the display module **30** in the third direction **Z** and formed of a material having higher conductivity than the side cover **90** to prevent current from flowing to, particularly, the side wire **46** according to an electrostatic discharge.

(290) Because the side wire **46** is positioned on the side surface **45** corresponding to each of the upper edge **32** and the lower edge **34** positioned in the third direction **Z**, as described above, the side member **100** may be positioned on an outer side of the side cover **90** formed in the third direction **Z** in which the side wire **46** is positioned.

(291) The side member **100** may easily guide static electricity to the metal plate **60** although the display modules **30A** to **30P** are not completely sealed due to a manufacturing failure.

(292) The side member **100** may cover the outer side of the side surface **45** of the substrate **40** in the third direction, as shown in FIG. 7. That is, the side member **100** may be positioned on a pair of edges **32** and **34** among the four edges **E** of the substrate **40**. More specifically, the side member **100** may be formed on one side surface **45** among side surfaces **45** corresponding to the pair of edges **32** and **34** on which the side wire **46** is positioned to reduce a width of a gap **G** formed between the display modules **30A** to **30P** upon an arrangement of the display modules **30A** to **30P**. This will be described in detail below.

(293) In regard of the first display module **30A**, the side member **100** may be positioned on the side surface **45** corresponding to the lower edge **34** of the first display module **30A**. Also, no side member **100** may be positioned on the side surface **45** corresponding to the upper edge **32** of the display module **30A**.

(294) In regard of the second display module **30B**, the side member **100** may be positioned only on the side surface **45** corresponding to the lower edge of the second display module **30B**, like the first display module **30A**, and in regard of the fourth display module **30C** and the sixth display module **30D**, the side member **100** may also be positioned only on the side surface **45** corresponding to the lower edge of each of the fourth display module **30C** and the sixth display module **30D**.

(295) The side member **100** may be formed of a metal material having higher conductivity than the side cover **90**. The side member **100** may be coated on the side cover **90** to be positioned on the side **91** of the side cover **90**.

(296) Accordingly, the side member **100** may be positioned in the gap **G** formed between the display modules **30A** to **30P** upon an arrangement of the display modules **30A** to **30P**.

(297) One end of the side member **100** may be in contact with the metal plate **60**, and the other end of the side member **100** may be positioned on the side **91** of the side cover **90**. That is, the side member **100** may cover at least one portion of the metal plate **60**, and at least one portion of the side cover **90** in the third direction **Z**.

(298) The side member **100** may be a thin film. The reason may be because the side member **100** is positioned in the gap **G** formed between the display modules **30A** to **30A** upon tiling of the display modules **30A** to **30P**. In a case in which the side member **100** is thick, the gap **G** formed between the display modules **30A** to **30P** may become wide due to the thickness of the side member **100**, which may cause recognition of a seam between the display modules **30A** to **30P**.

(299) However, although the side member **100** is a thin film, a certain spacing between neighboring ones of the display modules **30A** to **30P** may be made by the side member **100**. To reduce the certain spacing between the neighboring display modules **30A** to **30P**, a single side member **100** may be positioned between the sides **91** of the side covers **90** of the neighboring display modules

30A to 30P, which will be described in detail below.

(300) The side member **100** may be formed of a material having high conductivity. For example, the side member **100** may be formed of a metal, a conductive polymer, a conductive fabric, or the like, to be electrically grounded to the metal plate **60**.

(301) The side member **100** may be formed of a material having higher conductivity than the side cover **90**. Also, the side member **100** may be formed of a material having higher conductivity than the front cover **70**.

(302) Accordingly, current generated on the front cover **70** or the side cover **90** by an electrostatic discharge may not enter the substrate **40** because the current is not transmitted through the front cover **70** or the side cover **90**, and the current may enter the side member **100** while flowing on the front cover **70**.

(303) The current flowed to the side member **100** may flow to the metal plate **60** through the side member **100**, because the side member **100** is in contact with the metal plate **60** to be grounded to the ground component.

(304) That is, the side member **100** may provide a path of current, along which current generated on the front cover **70** or the side cover **90** by an electrostatic discharge flows to the metal plate **60** provided as a ground component. The side member **100** may guide charges generated by an electrostatic discharge to flow to the ground.

(305) Accordingly, because a major part of current generated on the front cover **70** or the side cover **90** by an electrostatic discharge flows to the metal plate **60** through the side member **100** having high conductivity, ESD pressure resistance of the electronic components mounted on the substrate **40** may be improved although some current flows to the substrate **40**.

(306) Additionally, electrostatic current transferred to the metal plate **60** may be discharged to an external ground through components, such as a bridge board, a cable, etc., which are in contact with the metal plate **60**.

(307) The side member **100** may have a dark color. The side member **100** may have a black color. The side member **100** may have a color that is darker than that of the front cover **70**.

(308) The side member **100** may have a color that is similar to that of the black matrix **48** or the side cover **90**. Accordingly, light entered the side member **100** may be absorbed in the side member **100** without being reflected.

(309) As described above, each of the display modules **30A to 30P** may include the front cover **70**, the side cover **90**, the metal plate **60**, and the side member **100**, independently, to prevent penetration of current according to an electrostatic discharge.

(310) As described above, the side member **100** may be positioned on only one side surface **45** of the pair of side surfaces **45** along which the side wire **46** extends.

(311) For example, in the first display module **30A** and the second display module **30B**, a side member **100A** may be positioned on the side surface **45** of the first display module **30A**, neighboring the second display module **30B** in the third direction **Z**.

(312) No side member may be positioned on the side surface **45** of the second display module **30B**, neighboring the first display module **30A** in the third direction **Z**. Accordingly, the side member **100A** of the first display module **30A** may face the side **91B** of the side cover **90** of the second display module **30B** in such a way as to be in contact with the side **91B**.

(313) Accordingly, a spacing **T** between the sides **91A** and **91B** of the first display module **30A** and the second display module **30B** may have a length corresponding to a thickness **t** of the side member **100**.

(314) In a case in which two side members **100** are positioned respectively on the pair of side surfaces **45** on which the side wire **46** is formed in the first and second display modules **30A** and **30B**, a spacing **T** between the sides **91A** and **91B** of the first and second display modules **30A** and **30B** may have a length corresponding to a total thickness **2t** of the two side members **100**. As a result, the spacing **T** may have a longer length to further increase the gap **G** between the first and

second display modules **30A** and **30B**, which raises recognition of a seam.

(315) However, according to an embodiment of the disclosure, only one side member **100A** may be positioned between the side **91A** of the first display module **30A** and the side **91B** of the second display module **30B**, as described above. Accordingly, the spacing **T** between the sides **91A** and **91B** of the first display module **30A** and the second display module **30B** may be reduced, resulting in a reduction of visibility of a seam between the display modules **30A** and **30B**.

(316) However, the side member **100** may be positioned on the side surface **45** corresponding to the upper edge **32**, instead of the side surface **45** corresponding to the lower edge **34**. In this case, the side member **100** may be positioned on the side surface **45** corresponding to the upper edge **32** of the second display module **30B** between the first display module **30A** and the second display module **30B**, and no side member **100** may be positioned on the first display module **30** between the first display module **30A** and the second display module **30B**.

(317) As such, although only one side member **100** of the first display module **30A** is positioned between the first display module **30A** and the second display module **30B**, current that is generated by an electrostatic discharge between the first display module **30A** and the second display module **30B** and enters the side cover **90B** of the second display module **30B** may be guided through the side member **100** of the first display module **30A**, being in contact with the side **91B** of the side cover **90B** of the second display module **30B**, resulting in high reliability against ESD of the second display module **30B**.

(318) That is, although the side member **100A** is positioned on the first display module **30A** between the first and second display modules **30A** and **30B** being adjacent to each other in the third direction **Z** among the display modules **30A** to **30P**, both the first and second display modules **30A** and **30B** may be in contact with the side member **100A**, and accordingly, reliability against ESD of the first and second display modules **30A** and **30B** may be raised through the side member **100A** in spite of current that is generated by an electrostatic discharge between the first and second display modules **30A** and **30B**.

(319) As described above, the first display module **30A** may be positioned at the uppermost location of the display panel **20**, and the side member **100A** may be positioned only on the side surface **45** corresponding to the lower edge **34** in the third direction **Z**. Accordingly, no side member **100** may be not positioned on the upper edge **32**.

(320) Also, in the second display module **30B**, the fourth display module **30C**, and the sixth display module **30D**, the side member **100** may be not positioned on the side surfaces **45** corresponding to the upper edges **32** of the second display module **30B**, the fourth display module **30C**, and the sixth display module **30D**, and side members **100B**, **100C**, and **100D** may be positioned respectively on the side surfaces **45** corresponding to the lower edges **34** of the second display module **30B**, the fourth display module **30C**, and the sixth display module **30D**.

(321) In this case, although the side members **100B**, **100C**, and **100D** of the second, fourth, and sixth display modules **30B**, **30C**, and **30D** are not positioned on the side surfaces **45** corresponding to the respective upper edges **32**, the side members **100A**, **100B**, and **100C** of the first, second, and fourth display modules **30A**, **30B**, and **30C** being adjacent to the second, fourth, and sixth display modules **30B**, **30C**, and **30D** in the third direction **Z** may be positioned on the side surfaces **45** corresponding to the upper edges **32** of the second, fourth, and sixth display modules **30B**, **30C**, and **30D**. Accordingly, reliability against ESD may be improved by the side members **100A**, **100B**, and **100C** of the adjacent first, second, and fourth display modules **30A**, **30B**, and **30C** although none of the side members **100B**, **100C**, and **100D** is positioned on the side surfaces **45** corresponding to the upper edges **32** of the second, fourth, and sixth display modules **30B**, **30C**, and **30D**.

(322) However, because there is no display module being adjacent to the side surface **45** corresponding to the upper edge **32** of the first display module **30A** in the third direction **Z**, current may enter the side surface **45** upon occurrence of an electrostatic discharge on the side surface **45**

corresponding to the upper edge **32** of the first display module **30A** to damage the side wire **46** and the electronic components of the first display module **30A**.

(323) To prevent such damage, the display apparatus **1** according to an embodiment of the disclosure may include a frame side member **200** positioned on the frame **15**, being in contact with the side surface **45** corresponding to the upper edge **32** of the first display module **30A** supported on the frame **15**, and grounded to the frame **15**.

(324) The frame **15** may be formed of a metal material and function as a ground component on a circuit.

(325) The frame side member **200** may be formed of a material having higher conductivity than the side cover **90**. For example, the frame side member **200** may be formed of a metal, a conductive polymer, a conductive fabric, or the like, to be electrically grounded to the metal plate **60**.

(326) The frame side member **200** may be formed at an area being adjacent to the upper edge of the frame **15** in the third direction **Z**. The frame side member **200** may extend in the second direction **Y** to cover all of the side surfaces **45** corresponding to the upper edges **32** of the third display module **30E**, the fifth display module **30I**, and the seventh display module **30M** aligned with the first display module **30A** in the second direction **Y**, as well as the first display module **30A**.

(327) That is, to protect the side surfaces **45** corresponding to the upper edges **32** of the third display module **30E**, the fifth display module **30I**, and the seventh display module **30M**, wherein no side member **100** is positioned on the side covers **90** formed on the side surfaces **45** corresponding to the upper edges **32**, as well as the first display module **30A**, the frame side member **200** may cover at least one portions of the side covers **90** of the first, third, fifth, and seventh display modules **30A**, **30E**, **30I**, and **30M**.

(328) Accordingly, upon applying of current caused by an electrostatic discharge onto the side surfaces **45** corresponding to the upper edges **32** of the first, third, fifth, and seventh display modules **30A**, **30E**, **30I**, and **30M**, the current may flow to the frame **15** through the frame side member **200**.

(329) Accordingly, although the side member **100** is positioned only on any one side **45** of the pair of side surfaces **45** on which the side wire **46** extends in each of the display modules **30A** to **30P**, as described above, ESD reliability of the other side surface **45** on which no side member **100** is positioned may be improved by the side member **100** of another display module being adjacent to the corresponding display module in the third direction **Z** or the frame side member **200**.

(330) Also, because the side member **100** is positioned only on any one side **45** of the pair of side surfaces **45** on which the side wire **46** extends in each of the display modules **30A** to **30P**, a length of a gap between display modules being adjacent to each other in the third direction **Z** may be reduced, resulting in a reduction of recognition of seams between the display modules **30A** to **30P**.

(331) However, in a case in which the side member **100** is positioned on the side surface **45** corresponding to the lower edge **34** of each of the display modules **30A** to **30P** in the third direction **Z**, instead of the upper edge **32**, the frame side member **200** may extend along the second direction at an area of the frame **15**, being adjacent to the lower edge **34**.

(332) In this case, the frame side member **200** may cover the side cover **90** formed on the side surface **45** corresponding to the lower edge **34** of the sixth display module **30D**, and the side covers **90** formed on the side surfaces **45** corresponding to the lower edges **34** of the display modules **30H**, **30L**, and **30P** aligned with the sixth display module **30D** in the second direction **Y**.

(333) Additionally, the side member **100** may be positioned only on any one side surface of a pair of side surfaces **45** on which no side wire **46** extends, that is, side surfaces **45** corresponding to the left edge **33** or the right edge **31** being opposite to each other in the second direction **Y**.

(334) The reason may be to prevent current from entering the side cover **90** formed on the side surface **45** corresponding to the left edge **33** or the right edge **31** to damage electronic components of the corresponding display module, although no side wire **46** is positioned on the side surface **45**.

(335) In this case, the side member **100** may be positioned on any one side surface **45** of the pair of

side surfaces in the second direction Y. On the side surface **45** on which the side portion **100** is not positioned in the second direction Y in each of the display modules **30A** to **30P**, a side member **100** of another display module being adjacent to the corresponding display module in the second direction Y may be positioned to raise ESD reliability.

(336) The side member **100** positioned in the second direction Z and the side member **100** positioned in the second direction Y may be integrated into one body, and accordingly, the side member **100** may have a shape surrounding two neighboring side surfaces **45** among four side surfaces **45** of each of the display modules **30A** to **30P**. For example, the side member **100** may surround the side covers **90** formed on the side surfaces **45** corresponding to the right edge **31** and the lower edge **34**.

(337) Also, in a case in which the side member **100** is additionally positioned on the side surface **45** corresponding to the right edge **31** of each of the display modules **30A** to **30P** in the second direction Y, the display apparatus **1** may further include the frame side member **200** extending in the third direction Z at an area being adjacent to the left edge of the frame **15**.

(338) Hereinafter, the display modules **30A** to **30P** according to another embodiment of the disclosure will be described. The remaining components except for a side wire **46** formed in the second direction Y, which will be described below, are the same as the corresponding ones of the display apparatus **1** according to an embodiment of the disclosure as described above, and therefore, overlapping descriptions thereof will be omitted. Particularly, an arrangement in third direction Z of the display modules **30A** to **30P** according to another embodiment of the disclosure is the same as that of the display apparatus **1** shown in FIGS. **6** and **7**, and therefore, an arrangement in second direction Y of the display modules **30A** to **30P** will be described.

(339) FIG. **9** is a perspective view showing some components of a display module according to another embodiment of the disclosure, FIG. **10** is a cross-sectional view showing some components of a display apparatus according to another embodiment of the disclosure in a second direction, FIG. **11** is a cross-sectional view of the some components shown in FIG. **10**, and FIG. **12** schematically shows a front side of some components of a display apparatus according to another embodiment of the disclosure.

(340) FIG. **9** shows a substrate **40** from which a component such as an anisotropic conductive layer **47** is excluded, for convenience of description. Also, a side wire **46** may include a coating member **46a** (see FIG. **7**) for protecting the side wire **46** from outside, and for convenience of description.

(341) As shown in FIG. **9**, the side wire **46** may be positioned on all four side surfaces **45** corresponding to four edges **31**, **32**, **33**, and **34** of a display module **30**.

(342) That is, according to the above-described embodiment of the disclosure, the side wire **46** may be positioned only on a pair of side surfaces **45** of the display module **30**. However, according to another embodiment of the disclosure, the side wire **46** may be formed on all the four side surfaces **45**.

(343) Accordingly, a side member **100** may be formed on one side surface **45** of a pair of side surfaces **45** being opposite to each other in the third direction Z and one side surface **45** of a pair of side surfaces **45** being opposite to each other in the second direction Y.

(344) The side member **100** may cover an outer side of a side surface **45** of a substrate **40** in the second direction Z, as shown in FIGS. **10** and **11**. More specifically, side members **100** may be positioned respectively on a side surface **45** corresponding to one edge **34** of two edges **32** and **34** positioned in the third direction Z among four edges E of the substrate **40** and a side surface **45** corresponding to one edge **31** of two edges **31** and **32** positioned in the second direction Y.

(345) The reason may be to reduce widths of gaps G formed between display modules **30A** to **30P** upon an arrangement of the display modules **30A** to **30P**.

(346) In regard of a first display module **30A**, the side member **100** may be positioned on a side surface **45** corresponding to a right edge **31** of the first display module **30A**. Also, the side member **100** may be not positioned on a side surface **45** corresponding to a left edge **33** of the display

module **30A**.

(347) In a third display module **30E**, the side member **100** may be positioned only on a side surface **45** corresponding to a right edge of the third display module **30E**, like the first display module **30E**, and in each of the fifth display module **30I** and the seventh display module **30M**, the side member **100** may also be positioned only on a side surface **45** corresponding to a right edge **31**.

(348) Accordingly, the side member **100** may be positioned in a gap G formed between the display modules **30A** to **30P** upon an arrangement of the display modules **30A** to **30P**.

(349) The side member **100** may cover at least one portion of a metal plate **60** and at least one portion of a side cover **90** in the second direction Y.

(350) For example, in the first display module **30A** and the third display module **30E**, a side member **100A** may be positioned on a side surface **45** of the first display module **30A**, the side surface **45** being adjacent to the third display module **30E** in the second direction Y.

(351) The side member **100A** may be not positioned on a side surface **45** of the third display module **30E**, the side surface **45** being adjacent to the first display module **30A** in the second direction Y. Accordingly, the side member **100A** of the first display module **30A** may face a side **91E** of a side cover **90** of the third display module **30E** in such a way as to be in contact with the side **91E**.

(352) Accordingly, a spacing T between sides **91A** and **91E** of the first display module **30A** and the third display module **30E** may have a length corresponding to a thickness t of the side member **100**.

(353) Accordingly, the spacing T between the sides **91A** and **91E** of the first display module **30A** and the third display module **30E** may be reduced, which leads to a reduction of recognition of a seam between the first display module **30A** and the third display module **30E**.

(354) However, the side member **100** may be positioned on the side surface **45** corresponding to the left edge **33**, not the side surface **45** corresponding to the right edge **31**. In this case, a side member **100** may be positioned on a side surface **45** corresponding to a left edge **33** of the third display module **30E** between the first display module **30A** and the third display module **30E**, and the side member **100A** of the first display module **30A** may be not positioned between the first display module **30A** and the third display module **30E**.

(355) As such, although only one side member **100A** of the first display module **30A** is positioned between the first display module **30A** and the third display module **30E**, current generated by an electrostatic discharge between the first display module **30A** and the third display module **30E** and entered the side cover **90E** of the third display module **30E** may be guided through the side member **100A** of the first display module **30A** being in contact with the side **91E** of the side cover **90E** of the third display module **30E**, resulting in high reliability against ESD of the third display module **30E**.

(356) That is, although the side member **100A** of the first display module **30A** is positioned between the first and third display modules **30A** and **30E** being adjacent to each other in the second direction Y among the display modules **30A** to **30P**, both the first and third adjacent display modules **30A** and **30E** may be in contact with the side member **100A**, and accordingly, reliability against ESD of both the first and third display modules **30A** and **30E** may be raised through the one side member **100A** in spite of current generated by an electrostatic discharge between the first and third display modules **30A** and **30E**.

(357) As described above, the first display module **30A** may be positioned at a leftmost location of the display panel **20**, and the side member **100A** may be positioned only on the side surface **45** corresponding to the right edge **31** in the second direction Y. Accordingly, no side member **100** may be positioned on the left edge **33**.

(358) Also, in the third display module **30E**, the fifth display module **30I**, and the seventh display module **30M**, the side member **100** may be not positioned on the side surfaces **45** corresponding to the left edges **33** of the third display module **30E**, the fifth display module **30I**, and the seventh display module **30M**, and side members **100E**, **100I**, and **100M** may be positioned respectively on

the side surfaces **45** corresponding to the right edges **31** of the third display module **30E**, the fifth display module **30I**, and the seventh display module **30M**.

(359) In this case, although the side members **100E**, **100I**, and **100M** of the third, fifth, and seventh display modules **30E**, **30I**, and **30M** are not positioned on the side surfaces **45** corresponding to the left edges **33**, the side members **100A**, **100E**, and **100I** of the first, third, and fifth display modules **30A**, **30E**, and **30I** being adjacent to the third, fifth, and seventh display modules **30E**, **30I**, and **30M** in the second direction **Y** may be positioned on the side surfaces **45** corresponding to the left edges **33** of the third, fifth, and seventh display modules **30E**, **30I**, and **30M**. Accordingly, reliability against ESD may be improved by the side members **100A**, **100E**, and **100I** of the adjacent first, third, and fifth display modules **30A**, **30E**, and **30I** although none of the side members **100E**, **100I**, and **100M** is positioned on the side surfaces **45** corresponding to the left edges **33** of the third, fifth, and seventh display modules **30E**, **30I**, and **30M**.

(360) However, because there is no display module being adjacent to the side surface **45** corresponding to the left edge **33** of the first display module **30A** in the second direction **Y**, current may enter the side surface **45** corresponding to the left edge **33** of the first display module **30A** upon occurrence of an electrostatic discharge on the side surface **45** to damage the side wire **46** and the electronic components of the first display module **30A**.

(361) To prevent such damage, the display apparatus **1** according to an embodiment of the disclosure may include a frame side member **200** including a first frame side member **210** positioned on a frame **15**, being in contact with the side surface **45** corresponding to the upper edge **32** of the first display module **30A** supported on the frame **15**, and grounded to the frame **15**, and a second frame side member **220** being in contact with the side surface **45** corresponding to the left edge **33** and grounded to the frame **15**.

(362) The frame side member **200** according to the above-described embodiment of the disclosure may extend in the second direction **Y** at an area being adjacent to the upper edge of the frame **15**, whereas, in the frame side member **200** according to another embodiment of the disclosure, the first frame side member **210** extending in the second direction **Y** at the area being adjacent to the upper edge of the frame **15** and the second frame side member **220** extending in the third direction **Z** at an area being adjacent to the left edge of the frame **15** may be positioned respectively on the frame **15**.

(363) The second frame side member **220** may extend in the third direction **Z** to cover all side surfaces **45** corresponding to left edges **33** of the second display module **30B**, the fourth display module **30C**, and the sixth display module **30D** aligned with the first display module **30A** in the third direction **Z**, as well as the first display module **30A**.

(364) That is, to protect the side surfaces **45** corresponding to the left edges **33** of the second display module **30B**, the fourth display module **30C**, and the sixth display module **30D** in which no side member **100** is positioned on the side covers **90** formed on the side surfaces **45** corresponding to the left edges **33**, as well as the first display module **30A**, the frame side member **200** may cover at least one portions of the side covers **90** of the first, second, fourth, and sixth display modules **30A**, **30B**, **30C**, and **30D**.

(365) Accordingly, as described above, although the side member **100** is positioned only on any one side surface **45** of a pair of side surfaces **45** on which the side wire **46** extends in each of the display modules **30A** to **30P**, ESD reliability of the other side surface **45** on which the side member **100** is not positioned may be improved by a side member **100** of another display module being adjacent to the corresponding display module in the second direction **Y** or the frame side member **200**.

(366) Also, because the side member **100** is positioned only on any one side **45** of the pair of side surfaces **45** on which the side wire **46** extends in each of the display modules **30A** to **30P**, a length of a gap **G** between display modules being adjacent to each other in the third direction **Z** may be reduced, resulting in a reduction of recognition of seams between the display modules **30A** to **30P**.

(367) However, in a case in which the side member **100** is positioned on the side surface **45**

corresponding to the left edge **33** of each of the display modules **30A** to **30P** in the second direction **Y**, instead of the right edge **31**, the frame side member **200** may extend along the third direction **Z** at an area of the frame **15**, being adjacent to the right edge **31**.

(368) In this case, the frame side member **200** may cover the side cover **90** formed on the side surface **45** corresponding to the right edge **31** of the seventh display module **30M**, and side covers **90** formed on side surfaces **45** corresponding to right edges **31** of display modules **30N**, **30O**, and **30P** aligned with the seventh display module **30M** in the third direction **Z**.

(369) The side member **100** positioned in the third direction **Z** and the side member **100** positioned in the second direction **Y** may be integrated into one body, and accordingly, the side member **100** may have a shape surrounding two neighboring side surfaces **45** among four side surfaces **45** of each of the display modules **30A** to **30P**.

(370) For example, in the first display module **30A**, the side member **100** may surround the side covers **90** formed on the side surfaces **45** corresponding to the right edge **31** and the lower edge **34**.

(371) The display apparatus according to an embodiment of the disclosure may be sealed by the front cover in the front direction of each display module, the side cover on the side surface of the display module, and the metal plate on the rear side of the display module, and ESD pressure resistance of the display apparatus may be improved by the side member additionally positioned on the side surface of the display module and grounded to the metal plate, against an electrostatic discharge that may occur in the display modules while the display modules are manufactured and conveyed and after the display modules are assembled into the display apparatus.

(372) The display apparatus according to an embodiment of the disclosure may have a seamless effect by which a seam between display modules is invisible by improving an arrangement of the side members of the display modules to reduce gaps between the display modules.

(373) So far, although the technical concept of the disclosure has been described based on specific embodiments, the scope of rights of the disclosure is not limited to these embodiments. It should be interpreted that various embodiments modified or changed by a person skilled in the art within a scope not deviating from the gist of the disclosure as the technical concept of the disclosure, which is defined in the claims, also belong to the scope of rights of the disclosure.

Claims

1. A display module comprising: a substrate having a mounting surface on which a plurality of inorganic light-emitting diodes are mounted and a TFT layer is formed, four side surfaces, and a rear surface being opposite to the mounting surface; a side wiring electrically connected to the TFT layer and extending along a first pair of side surfaces among the four side surfaces of the substrate; a front cover covering the TFT layer and the plurality of inorganic light emitting devices in a first direction; a metal plate provided on the rear surface of the substrate; a side cover covering the side wiring and the four side surfaces; and a side member provided on a side of the side cover and grounded to the metal plate, wherein the side member is provided on a first side surface of the first pair of side surfaces along which the side wiring extends among the four side surfaces.
2. The display module of claim 1, wherein the side member comprises a first side member provided on the first side surface of the first pair of side surfaces along which the side wiring extends, and a second side member provided on a second side surface of a second pair of side surface along which the side wiring does not extend among the four side surfaces.
3. The display module of claim 2, wherein the first side member and the second side member are integrally formed as one body and extend along the first side surface and the second side surface.
4. The display module of claim 1, wherein the side wiring extends along the four side surfaces, and wherein the side member is provided on the first side surface of the first pair of side surfaces and a second side surface of a second pair of side surfaces among the four side surfaces.
5. The display module of claim 4, wherein the side member positioned on the first side surface and

the second side surface is integrally formed as one body.

6. The display module of claim 1, wherein a conductivity of the side member is greater than a conductivity of the side cover.

7. The display module of claim 1, wherein a color of the side member is black.

8. The display module of claim 1, wherein the side member comprises a metal material.

9. A display apparatus comprising: a plurality of display modules are arranged in an M*N matrix form; and a frame configured to support the plurality of display modules, wherein each display module of the plurality of display modules comprises: a substrate having a mounting surface on which a plurality of inorganic light-emitting diodes are mounted and a TFT layer is formed, four side surfaces, and a rear surface being opposite to the mounting surface; a side wiring electrically connected with the TFT layer and extending along a first pair of side surfaces among the four side surfaces; a front cover covering the plurality of inorganic light-emitting diodes and the TFT layer in a first direction; a metal plate provided on the rear surface; a side cover covering the side wiring and the four side surfaces; and a side member provided on a side of the side cover and grounded to the metal plate, and wherein the side member is positioned on a first side surface of the first pair of side surfaces along which the side wiring extends among the four side surfaces.

10. The display apparatus of claim 9, wherein the plurality of display modules comprises a first display module and a second display module that is adjacent to the first display module in a direction in which the side wiring of the first display module is extends, and wherein the side member of the first display module contacts a second side surface of the second display module among the first pair of side surfaces along which the side wiring of the second display module extends, and the side member of the second display module is not provided on the second side surface of the second display module.

11. The display apparatus of claim 10, wherein a first display module of the plurality of display modules is positioned such that a second side surface among the first pair of side surfaces along which the side wiring extends is adjacent to an edge of the frame, and the side member is not positioned on the second side surface, and wherein the frame comprises a frame side member that surrounds the second side surface of the first display module among the first pair of side surfaces along which the side wiring extends.

12. The display apparatus of claim 11, wherein the frame side member extends along the edge of the frame.

13. The display apparatus of claim 9, wherein the side member comprises a first side member provided on the first side surface of the first pair of side surfaces along which the side wiring extends, and a second side member provided on a second side surface of a second pair of side surface along which the side wiring does not extend among the four side surfaces.

14. The display apparatus of claim 13, wherein the first side member and the second side member are integrally formed as one body and extend along the first side surface and the second side.

15. The display apparatus of claim 13, wherein each display module of the plurality of display modules is positioned such that side surfaces on which neither the first side member nor the second side member are positioned, among the four side surfaces, are respectively adjacent to edges of the frame, and wherein the frame comprises a frame side member that surrounds the side surfaces of the each display module of the plurality of display modules, on which neither the first side member nor the second side member are provided.
